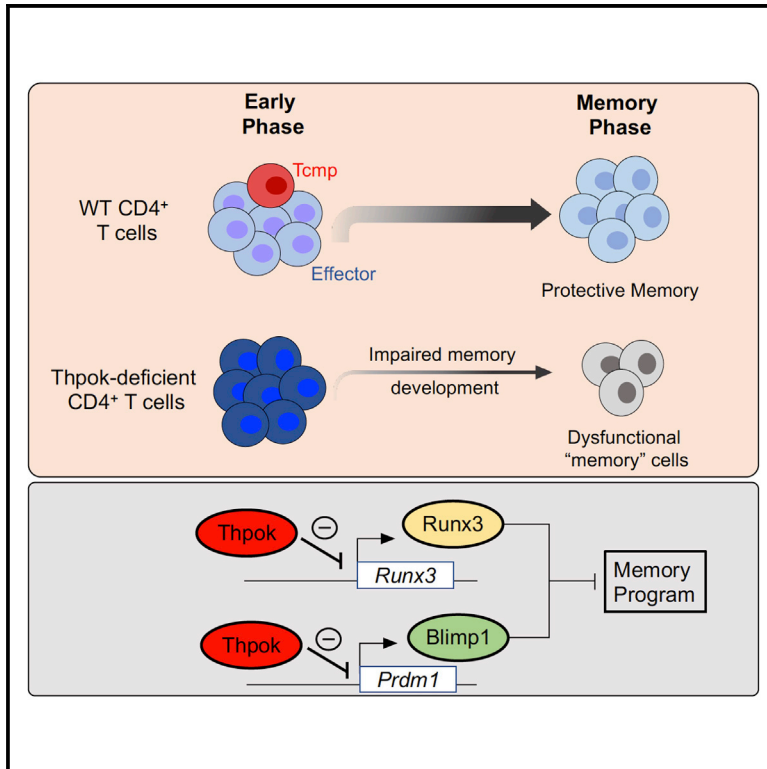


The Emergence and Functional Fitness of Memory CD4⁺ T Cells Require the Transcription Factor Thpok

Graphical Abstract



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In Brief

Memory CD4⁺ T cells contribute to long-term immunity against pathogens. Ciucci et al. show that the transcription factor Thpok is essential to safeguard the memory potential of early pathogen-responding CD4⁺ T cells and serves by antagonizing a Blimp1- and Runx3-driven dysfunction transcriptional circuitry.

Highlights

- Single-cell RNA sequencing characterizes CD4⁺ central memory precursor T cells
- CD4⁺ central memory precursor T cells depend on the transcription factor Thpok
- Protective memory responses are impaired by disruption of Thpok in CD4⁺ T cells
- Thpok represses a dysfunction, exhaustion-like program driven by Blimp1 and Runx3



The Emergence and Functional Fitness of Memory CD4⁺ T Cells Require the Transcription Factor Thpok

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SUMMARY

Memory CD4⁺ T cells mediate long-term immunity, and their generation is a key objective of vaccination strategies. However, the transcriptional circuitry controlling the emergence of memory cells from early CD4⁺ antigen-responders remains poorly understood. Here, using single-cell RNA-seq to study the transcriptome of virus-specific CD4⁺ T cells, we identified a gene signature that distinguishes potential memory precursors from effector cells. We found that both that signature and the emergence of memory CD4⁺ T cells required the transcription factor Thpok. We further demonstrated that Thpok cell-intrinsically protected memory cells from a dysfunctional, effector-like transcriptional program, similar to but distinct from the exhaustion pattern of cells responding to chronic infection. Mechanistically, Thpok-bound genes encoding the transcription factors Blimp1 and Runx3 and acted by antagonizing their expression. Thus, a Thpok-dependent circuitry promotes both memory CD4⁺ T cells' differentiation and functional fitness, two previously unconnected critical attributes of adaptive immunity.

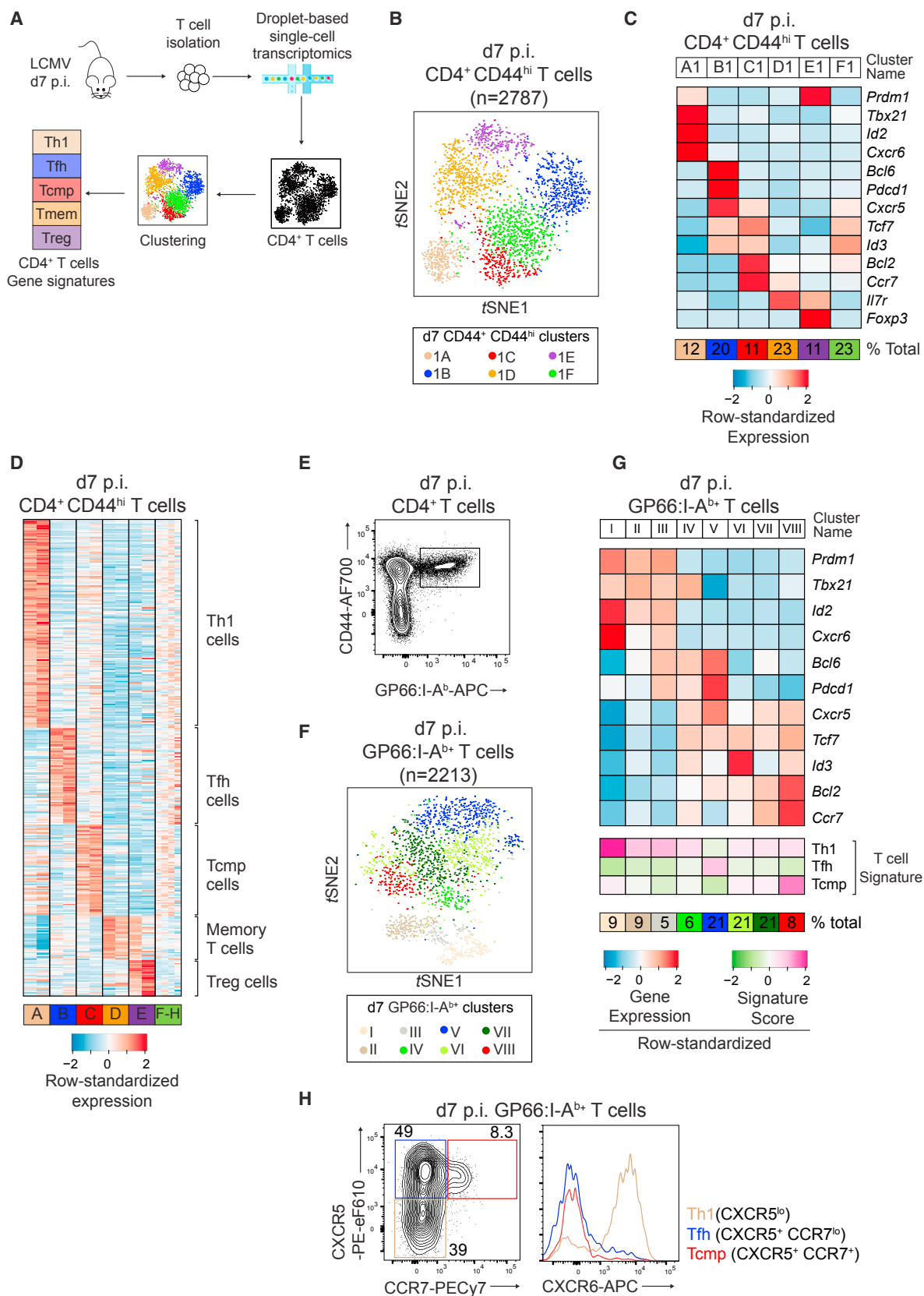
INTRODUCTION

Immunological memory, the ability to mount a fast secondary effector response to a new challenge by a previously encountered pathogen, is a defining property of the adaptive immune system, and it relies on long-lived antigen-specific B and T cells. T cells responding to an initial infection undergo extensive proliferation, giving rise to effector populations that promote pathogen clearance. After a contraction phase when most effector T cells die by apoptosis, memory T cells survive and provide long-lasting protection against reinfection.

In addition to promoting CD8⁺ T cell and B cell memory responses, CD4⁺ T cells give rise to memory populations essential for durable immunity (Laidlaw et al., 2016; MacLeod et al., 2009), including against human influenza, malaria, and Leishmania infection (Mordmüller et al., 2017; Wilkinson et al., 2012; Zaph et al., 2004). Although most antigen-responding CD4⁺ T cells have the potential to give rise to short-lived effector and long-lived memory cells (Cho et al., 2017; Tubo et al., 2016), what controls the emergence and functional fitness of memory CD4⁺ cells remains poorly understood. Unlike for memory CD8⁺ T cells, whose precursors can be identified early during the immune response (e.g., through their expression of interleukin-7 receptor [IL-7R α]), identifying putative precursors of memory CD4⁺ T cells has been challenging (Ahmed et al., 2009; Marshall et al., 2011; Pepper and Jenkins, 2011). One perspective holds that "memory precursors" diverge from short-term effectors early during the immune response, notably by restraining expression of differentiation programs (including interferon- γ [IFN γ]-expressing T helper-1 [Th1] cells) specific to effector fates and maintaining expression of genes characteristic of undifferentiated cells; supporting this idea, studies have identified specific CD4⁺ T cell subsets that exhibit higher potential for memory differentiation (Cho et al., 2017; Lüthje et al., 2012; Marshall et al., 2011; Nish et al., 2017; Pepper et al., 2011). However, there is support for the possibility that differentiated effectors can give rise to long-lived memory cells (Fazilleau et al., 2009; Harrington et al., 2008; Marshall et al., 2011; McKinstry et al., 2014).

Here, we have taken advantage of single-cell RNA sequencing (scRNA-seq) to characterize the gene expression programs underpinning the heterogeneity of T cell responses and the development of memory T cells. Analyzing CD4⁺ T cells at the peak of their response to an acute viral infection, we identified a transcriptional signature characteristic of potential memory-cell precursors. We found that both the emergence of this signature and the generation of a durable and effective memory CD4⁺ response required the transcription factor Thpok, which notably prevented the establishment of a dysfunction gene-expression pattern characteristic of short-lived effector cells. Mechanistically, Thpok protected memory development by restraining the





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expression of the transcription factors Runx3 and Blimp1 and bound to chromatin regions within or near these genes. Thus, a Thpok-dependent transcriptional circuitry controls the emergence of a memory program in early CD4⁺ T cell responders, and it is critical for the establishment of durable and effective CD4⁺ T cell memory.

RESULTS

Single-Cell RNA-Seq Identifies the Signature of a CD4⁺ T Cell Memory Precursor

We performed scRNA-seq to dissect early T cell responses against the Armstrong strain of Lymphocytic Choriomeningitis Virus (LCMV Arm), which in wild-type (WT) mice is cleared by innate and CD8⁺ T cells (Matloubian et al., 1994). At 7 days post-infection (dpi), we used the droplet-based GemCode Chromium technology (Figure 1A) (Zheng et al., 2017) to capture non-naïve (CD44^{hi}) spleen T cells, which included LCMV-reactive cells and pre-existing memory cells of unrelated specificity. We first analyzed 7,006 cells, each expressing an average of 1,446 genes (Table S1). Visualization with t-distributed stochastic neighbor embedding (t-SNE), which positions cells on the basis of similarity in gene expression, revealed a clear demarcation between CD4⁺ and CD8⁺ T cells, as well as negligible contamination by non-T cells (Figure S1A). Subsequent analyses were carried out on CD4⁺ T cells only and were aimed at identifying cell subsets defined by specific gene expression signatures. Unsupervised clustering of CD44^{hi} CD4⁺ T cells (2,782 cells) identified 6 clusters (clusters A1–F1) matching the known heterogeneity of this population (Hale et al., 2013; Marshall et al., 2011) (Figures 1B and S1B). Clusters A1, B1, and E1 included cells expressing *Tbx21* (encoding T-bet, characteristic of Th1 cells), *Bcl6* (T follicular helper [Tfh] cells), and *Foxp3* (regulatory T [Treg] cells) (Figure 1C). Cluster D1 cells expressed genes, including *Irf7*, *Bcl2*, and *Ccr7*, that are typical of memory cells, (Kaech et al., 2002; Marshall et al., 2011). Cluster D1's high expression of *Irf7*, a gene minimally expressed at 7 dpi in virus-specific CD4⁺ T cells (Purton et al., 2007), suggested that it mostly included pre-existing memory cells (Tmem) unrelated to LCMV infection. For each cluster, we computed signatures based on genes differentially expressed between that cluster and all others.

Using the same approach on an independent replicate of 956 CD44^{hi} CD4⁺ spleen T cells at 7 dpi (Table S1), we identified eight clusters (Figure S1C); five of these (A2–E2) had gene expression signatures closely matching those that were identified for clusters A1–E1 in the first replicate, highlighting the robustness of this anal-

ysis. There was less similarity between clusters F1 and F2–H2, which could not be assigned a specific signature (Figure S1D). Thus, we defined Th1, Tfh, Treg, and Tmem cell signatures as the intersection of corresponding gene signatures defined in each replicate (Figure 1D and Table S2). In addition, the signature defined by clusters C1 and C2 comprised genes (including *Tcf7* [encoding TCF-1], *Bcl2* or *Ccr7*, but not *Irf7*) typical of memory cells, and it excluded typical Tfh and Th1 cell effector genes. Because CCR7-expressing CD4⁺ T cell responders have an increased potential to become memory cells (Cho et al., 2017; He et al., 2013; Marshall et al., 2011; Pepper and Jenkins, 2011; Sallusto et al., 1999), we speculated that this signature, which we designated T central memory precursor (Tcmp), identified a gene expression program related to memory potential.

To unambiguously assess virus-specific cells, we performed scRNA-seq at 7 dpi on CD4⁺ spleen T cells specific for the LCMV glycoprotein-derived GP66 peptide complexed to I-A^b MHC-II molecules (Figure 1E and Table S1). We clustered this LCMV-specific population and scored CD4⁺ T cells in every cluster for each gene signature we had previously defined (Figures 1F and 1G). GP66-specific cells were separated into clusters with high scores for Th1 (clusters I–III), Tfh (cluster V), and Tcmp (cluster VIII) cell signatures (Figures 1G and S1E). In contrast, GP66-specific cells showed little or no enrichment for the Treg cell signature, consistent with the lack of a *Foxp3*⁺ Treg cell component in the GP66-specific response (Crawford et al., 2014) (Figures S1F and S1G). GP66-specific cells also scored low for the Tmem cell signature (Figure S1H), in agreement with the idea that this signature identifies memory or memory-like cells that were not reactive against a pre-existing infection with LCMV. Additional clusters partially matched these well-defined signatures but were not robustly identified across experiments (Figures 1D and 1G), suggesting functional plasticity among early CD4⁺ T cell responders. However, high scores for Tfh and Tcmp cell signatures were mutually exclusive (Figures 1D, 1G, S1E, and S1I) even though expression of specific genes (e.g., *Cxcr5* and *Tcf7*) was observed in clusters corresponding to each signature. Taken together, these scRNA-seq analyses identified a specific Tcmp cell transcriptional signature, distinct from that of differentiated effector cells and highly enriched in a CCR7-expressing subset known to have elevated CD4⁺ T cell memory potential. The demarcation between Tfh and Tcmp cell signatures matched the separation between CCR7[−] CXCR5⁺ (Tfh) and CCR7⁺ CXCR5⁺ cells in flow cytometry analyses (Figure 1H); thus, subsequent experiments used co-expression of both receptors to identify cells with elevated memory potential. Of note, some Th1 cells had positive scores

Figure 1. Identification of CD4⁺ T Cell Effector Signatures by scRNA-Seq

- (A) Schematic of experimental procedures.
 (B–D) scRNA-seq of wild-type (WT) CD4⁺ CD44^{hi} T cells at 7 days post infection (dpi) with the Armstrong strain of lymphocytic choriomeningitis virus (LCMV Arm).
 (B) t-SNE plot (experiment 1); each dot is a CD4⁺ T cell, and colors highlight cell clusters.
 (C) Differential expression of select genes in clusters shown in (B).
 (D) Expression of signature components among CD4⁺ T cell clusters, either defined in both replicates (A–E; replicate 1 is on the left of each pair) or unique to each replicate (F–H).
 (E–G) scRNA-seq of WT GP66:I-A^b CD4⁺ T cells at 7 dpi.
 (E) Purification strategy.
 (F) t-SNE plot; colors highlight the eight clusters.
 (G) Expression of selected genes and signature scores among clusters shown in (F).
 (H) CCR7 versus CXCR5 expression on spleen T cells (left) defines Th1, T_{FH}, and Tcmp cell subsets, which are analyzed for CXCR6 expression (right). Data are from one representative experiment out of three. Please see also Figure S1.

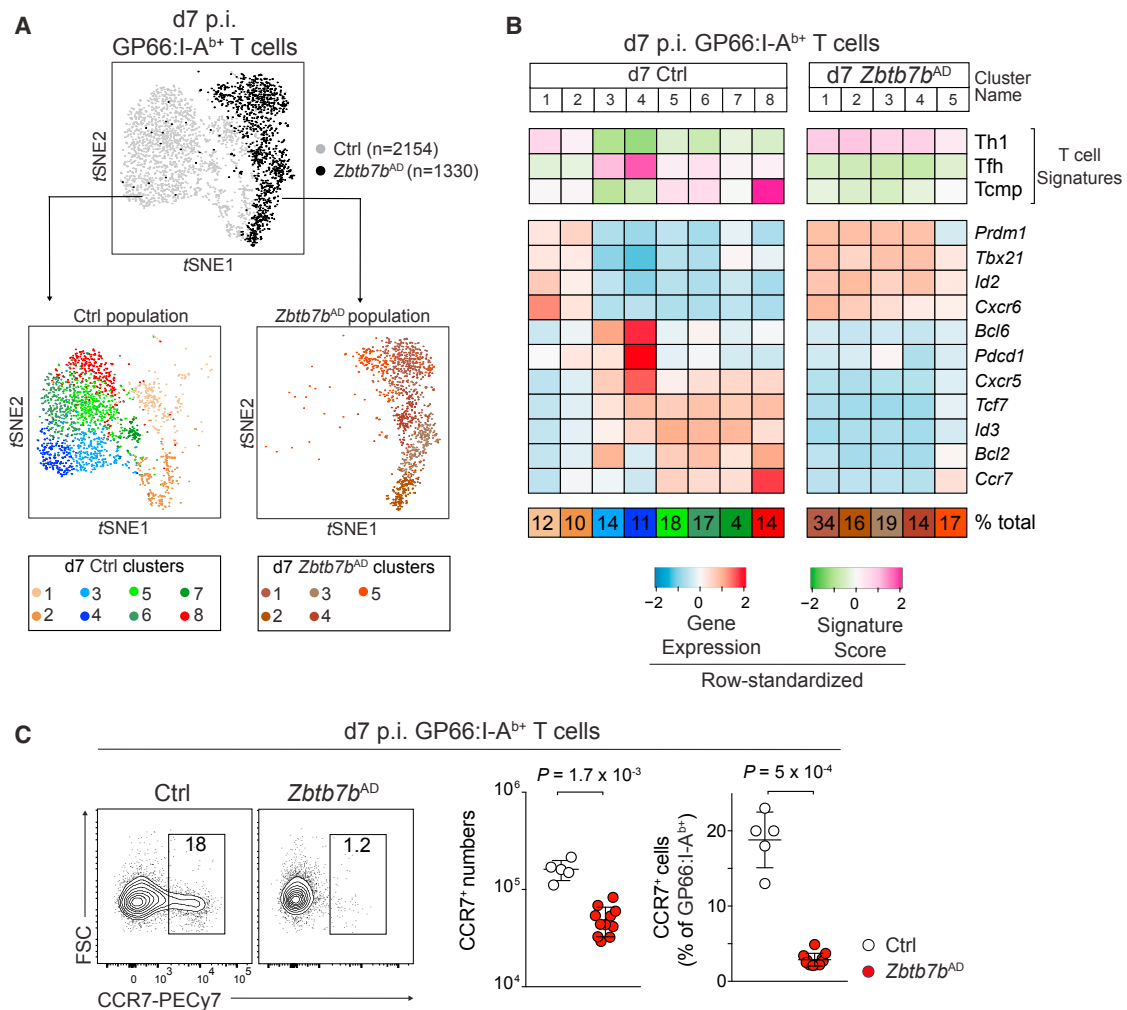


Figure 2. Thpok Controls the Emergence of the Tcmp Signature

(A and B) scRNA-seq of *Zbtb7b*^{AD} or control (*Tnfrsf4-cre*⁺ *Zbtb7b*^{+/+}) GP66:I-A^{b+} *Rosa26*^{YFP} spleen T cells at 7 dpi with LCMV Arm.

(A) The top t-SNE plot shows cells colored by genotype. The bottom t-SNE plots show cells from each genotype, independently clustered and color-coded by cluster.

(B) Signature scores (top three rows) and expression of selected genes (middle panel) among clusters of *Zbtb7b*^{AD} and control cells displayed in (A). Percentages (bottom row) indicate cluster size relative to the total population.

(C) Contour plots (left) and quantification (right) of CCR7 expression on spleen GP66:I-A^{b+} T cells. Data are from 11 *Zbtb7b*^{AD} mice and 5 control (Ctrl) mice; data were obtained in two experiments representative of three. Please see also Figure S2.

Error bars indicate standard deviation.

for genes of the Tcmp cell signature (Figures S1D and S1I), highlighting the fact that the expression of Th1 cell-associated genes does not exclude conservation of memory potential, as previously noted (Harrington et al., 2008; Marshall et al., 2011; Nish et al., 2017; Pepper et al., 2011; Tubo et al., 2016).

The Tcmp Cell Signature Is Thpok Dependent

We sought genetic evidence to support the functional significance of the Tcmp cell signature. We noted that there are substantial differences between CD4⁺ and CD8⁺ memory T cells (Crawford et al., 2014; Homann et al., 2001; Schiemann et al., 2003); accordingly, no 7 dpi LCMV CD8⁺ T cell cluster scored high for the Tcmp cell signature (data not shown). Because the transcription factor Thpok promotes CD4⁺ T cell differentiation in the thymus and because its

expression remains largely specific of CD4⁺ T cells after thymic egress (Carpenter and Bosselut, 2010; He et al., 2005; Sun et al., 2005), we inquired whether Thpok was necessary for the expression of the Tcmp cell signature. We analyzed the single-cell transcriptome of LCMV-specific CD4⁺ T cells deficient for *Zbtb7b* (encoding Thpok) by using *Zbtb7b*^{fl/fl} mice carrying a *cre* cDNA that was genetically targeted into *Tnfrsf4* (encoding Ox40) and designated *Zbtb7b*^{AD} (for “activation-induced deletion”). In these mice, Cre was preferentially expressed in antigen-activated post-thymic CD4⁺ T cells (Figure S2A), therefore leaving naive conventional CD4⁺ T cells untouched. These mice carried a *Rosa26*^{YFP} allele to identify Cre-expressing cells (Srinivas et al., 2001), and cells subject to scRNA-seq were sorted for yellow fluorescent protein (YFP) expression.

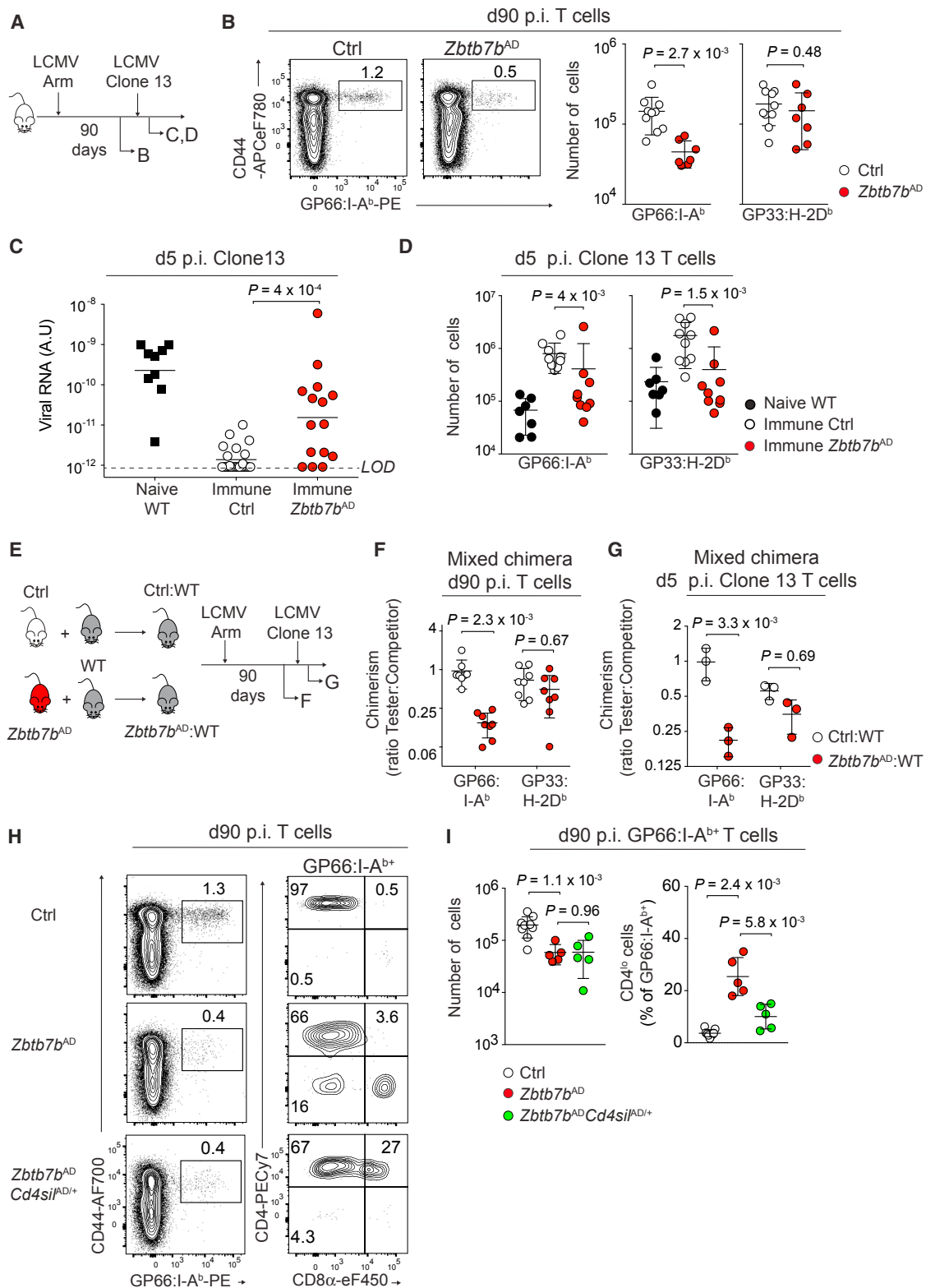


Figure 3. Thpok Promotes the Development of Memory CD4⁺ T Cells

(A) Schematic of the experiment shown in (B–D).

(B) GP66:I-A^b versus CD44 expression in spleen T cells at 90 dpi with LCMV Armstrong (left). Numbers of GP66:I-A^b+ and GP33:H-2D^b+ spleen T cells from 7 *Zbtb7b*^{AD} and 10 control mice (right). Data are from one experiment representative of three independent experiments.

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Thpok deletion in *Zbtb7b*^{AD} mice affected neither viral clearance nor expansion of antigen-specific cells (Figures S2B–S2E). However, t-SNE analyses of scRNA-seq data obtained from 1,330 *Zbtb7b*^{AD} and 2,154 control GP66-specific T cells at 7 dpi showed minimal overlap between the two genotypes (Figure 2A). Unsupervised clustering at 7 dpi identified clusters with high scores for Th1-, Tfh-, and Tcmp cell signatures among control cells (control clusters Ctrl1–2, Ctrl3–4, and Ctrl8, respectively). In contrast, the Thpok-deficient response at 7 dpi was dominated by clusters with a Th1 cell signature (clusters *Zbtb7b*^{AD}1–4, Figure 2B), in line with the high frequency of IFN γ -producing cells in response to GP66 peptide stimulation (Figure S2F). All Thpok-deficient clusters showed low scores for the Tcmp cell signature: although cluster *Zbtb7b*^{AD} 5 showed limited expression of *Bcl2* and *Ccr7*, it differed from the control Tcmp cell cluster Ctrl8 by its low expression of *Tcf7* and *Id3* and high expression of Th1 cell-associated genes *Id2* or *Tbx21*. In agreement with these results, the number of CCR7-expressing cells was greatly reduced in *Zbtb7b*^{AD} compared to control mice at 7 dpi (Figure 2C). Finally, the Thpok-deficient response had no Tfh component, suggesting that Thpok was needed for Tfh cell differentiation. We confirmed this in separate experiments (unpublished data). Thus, Thpok is necessary for the emergence of the Tcmp cell gene signature.

Cell-Intrinsic Thpok Is Required for the Generation of CD4⁺ Memory T Cells

The absence of cells with a high Tcmp signature score in Thpok-deficient mice prompted us to analyze the role of Thpok in memory CD4⁺ T cell development and maintenance. WT mice clear LCMV Arm by 7 or 8 dpi and generate populations of virus-specific CD4⁺ and CD8⁺ memory T cells (Matloubian et al., 1994). Analyses performed 90 days after LCMV Arm infection showed lower numbers of CD4⁺ but not CD8⁺ T cells in *Zbtb7b*^{AD} than in control animals (we evaluated the CD8⁺ T cell response by staining for the virus-derived GP33 peptide complexed to H-2D^b MHC-I molecules; Figures 3A and 3B). This suggested that Thpok was needed for CD4⁺ T cell memory. Normally, LCMV Arm infection results in durable LCMV immunity, including against the clone 13 strain that establishes chronic infection in non-immune mice. To evaluate the role of Thpok in immune memory, we compared the responses of *Zbtb7b*^{AD} and control mice to re-challenge by LCMV clone 13 more than 90 days after

infection with LCMV Arm. Thpok-deficient mice re-challenged with clone 13 showed impaired viral clearance (Figure 3C), similar to the impaired clearance in MHC II-deficient mice that lack CD4⁺ T cells (Figure S2G), and they had reduced numbers of both CD4⁺ and CD8⁺ LCMV-responsive T cells (Figure 3D). Although germline *Zbtb7b* disruption had been reported to directly affect CD8⁺ T cell memory responses (Setoguchi et al., 2009), it was unlikely that the effect on CD8⁺ T cells from *Zbtb7b*^{AD} mice was cell intrinsic because only 25%–30% of them expressed Cre (Figure S2C). To address this, we generated mixed bone-marrow chimeras combining allelically marked “tester” (either *Zbtb7b*^{AD} or control [*Zbtb7b*^{+/+} *Tnfrsf4-cre*]) and WT competitor components (Figure 3E). *Zbtb7b*^{AD} CD4⁺ T cells were out-competed by their WT counterparts at 90 dpi (Figure 3F) and remained so when chimeric mice were re-challenged with LCMV clone 13 (Figure 3G); this indicates a cell-intrinsic requirement for the generation of memory CD4⁺ T cells. In contrast, *Zbtb7b*^{AD} CD8⁺ T cells competed normally with WT cells; this indicated that, in addition to its cell-intrinsic effect on CD4⁺ T cell memory, Thpok is necessary for CD4⁺ T cell help with CD8⁺ T cell responses (Laidlaw et al., 2016).

We next analyzed the Thpok-deficient, antigen-specific “memory” cells that persisted several weeks after resolution of LCMV Arm infection. These cells had reduced CD4 surface expression (Figure S3A), consistent with the way Thpok promotes *Cd4* expression by antagonizing the *Cd4* transcriptional silencer (Muroi et al., 2008; Wildt et al., 2007). Indeed, disrupting the *Cd4* silencer in *Zbtb7b*^{AD} cells efficiently sustained *Cd4* expression; however, it failed to reconstitute normal memory populations (Figures 3H and 3I). Thus, although CD4 expression is important for CD4⁺ T cell homeostasis *in vivo* (Wang et al., 2001), the impact of Thpok on memory CD4⁺ T cell generation is independent of its effect on CD4 expression. We also noted that the GP66-specific Thpok-deficient population included an IL-7R α ^{lo} KLRG1^{hi} component typical of short-lived terminally differentiated effectors (Kaeche et al., 2003) (Figure 4A). This skewing toward effector differentiation was cell-intrinsic; it was observed both in mixed bone-marrow chimeras and in *Zbtb7b*^{AD} CD4⁺ T cells that expressed the I-A^b-GP66-specific SMARTA TCR transgene and that were adoptively transferred into LCMV-infected WT hosts (Figure S3B). It was not restored by enforced CD4 expression in *Zbtb7b*^{AD} *Cd4sil*^{AD} cells (Figure S3C). Additionally, skewing toward an IL-7R α ^{lo} KLRG1^{hi} phenotype

(C) Serum virus titers 5 days after LCMV clone 13 re-challenge of LCMV-Arm-immunized mice (except black squares: naive WT mice 5 dpi with clone 13 as controls). Data summarize four independent experiments; each symbol represents a separate mouse (*Zbtb7b*^{AD}, *n* = 15; Ctrl, *n* = 23; naive WT, *n* = 9). Bars denote the geometric mean; LOD = limit of detection.

(D) GP66:I-A^b and GP33:H-2D^b T cell numbers in immune *Zbtb7b*^{AD} (*n* = 9) or control (*n* = 10) mice and in naive WT mice (*n* = 7) at 5 dpi with LCMV clone 13. Data summarize two independent experiments.

(E) Schematic depiction of the experiment shown in (F and G). Mixed-bone-marrow chimeras made from WT ‘competitor’ and either *Zbtb7b*^{AD} or Ctrl ‘tester’ components were infected with LCMV Arm and either analyzed at 90 dpi (F) or infected with LCMV clone 13 90 days after LCMV Arm infection and analyzed 5 days after LCMV clone 13 infection (G). Each panel shows one out of two independent experiments.

(F) Tester:competitor ratios among GP66:I-A^b and GP33:H-2D^b spleen T cells in *Zbtb7b*^{AD}:WT (red) and Ctrl:WT (empty circles) chimeras (each symbol represents a distinct chimera, *n* = 8 per group).

(G) Tester:competitor ratios shown as in (F) at 5 days after LCMV clone 13 re-challenge (*n* = 3 per group). *p* values are from two-way ANOVA.

(H and I) Spleen T cells from indicated mice, analyzed at 90 dpi with LCMV Arm.

(H) CD8 α versus CD4 expression (right) gated on GP66:I-A^b cells (left plots).

(I) Numbers of GP66:I-A^b cells (left) and percentage of CD4^{lo} cells (right) in *Zbtb7b*^{AD} (*n* = 5), *Zbtb7b*^{AD} *Cd4sil*^{AD} (*n* = 5) and control (*n* = 9) animals. Data are from one out of two independent experiments. Please see also Figures S2 and S3. Error bars indicate standard deviation.

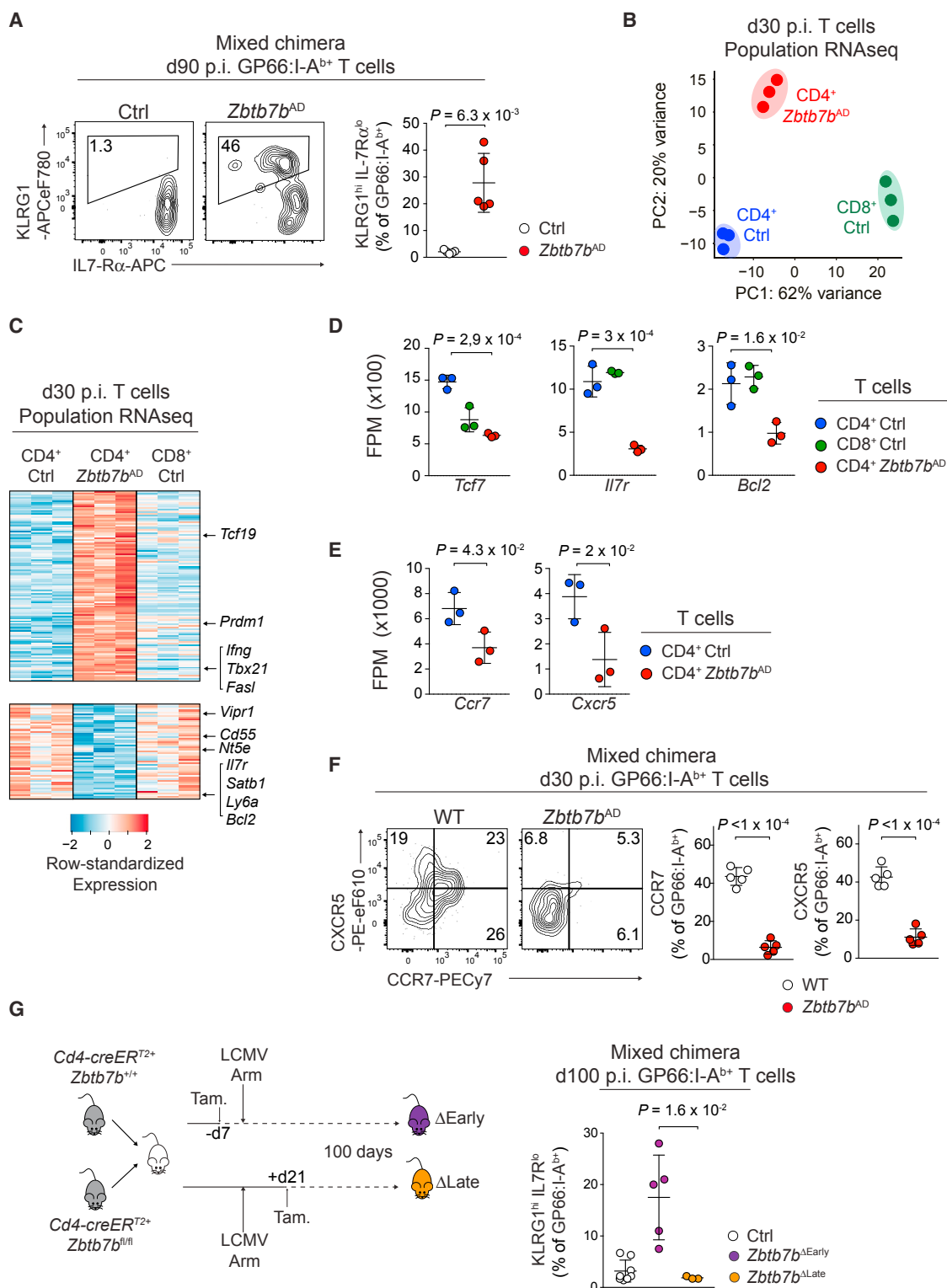


Figure 4. Early Impact of Thpok on CD4⁺ T Cell Memory Differentiation

(A) Contour plots (left) and quantification (right, $n = 5$ per group) of IL-7R α and KLRG1 expression in GP66:I-A^{b+} spleen T cells from *Zbtb7b*^{AD}:WT or Ctrl:WT mixed bone-marrow chimeras at 90 dpi. with LCMV Arm. Data are from one representative experiment out of two.

(B–E) RNA-seq analysis of gene expression in GP66:I-A^{b+} *Zbtb7b*^{AD} (CD4 *Zbtb7b*^{AD}) or control (CD4 Ctrl) and GP33:H-2D^{b+} (CD8 Ctrl) spleen T cells at 30 dpi with LCMV Arm.

(F) Principal-component analyses (PCA).

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was amplified during the recall response to clone 13 re-challenge (Figure S3D); this indicated that *Thpok* disruption not only reduced the number of memory T cells but also impaired their function on a single-cell basis.

To further analyze the impact of *Thpok* on CD4⁺ T cell memory, we performed RNA-seq on LCMV-specific T cell populations sorted 30 dpi with LCMV Arm, at which time the number of antigen-specific T cells was already lower in *Zbtb7b*^{AD} than in control mice (Figure S3E). Principal-component analyses showed that, at 30 dpi, *Thpok*-deficient CD4⁺ T cells differed from both CD4⁺ and CD8⁺ control memory T cells (Figure 4B). In addition to expressing CD8-lineage genes, consistent with previous reports (Mucida et al., 2013; Vacchio et al., 2014; Wang et al., 2008a) (Figures S3F and S3G), *Thpok*-deficient cells showed impaired expression of prototypical memory genes, including *Ccr7*, *Cxcr5*, *Tcf7*, *Bcl2*, or *Irf7* (Hale et al., 2013; Marshall et al., 2011) (Figures 4C–4E), and of CCR7 and CXCR5 proteins (Figure 4F). Conversely, the *Thpok*-deficient cells inappropriately expressed genes, including *Ifng*, *Tbx21*, *Id2*, or *Prdm1* (encoding the transcription factor Blimp1), that are typical of Th1 cells (Figure S3H). Consistent with this Th1 cell effector diversion, *Thpok*-deficient cells had increased expression of Granzyme B (Figures S3H and S3I). Thus, *Thpok* is necessary for the appropriate expression of a memory program in the CD4⁺ T cells that persist after resolution of infection.

Thpok Is Required Early During Activation for the Emergence of the Memory Program

Because scRNA-seq analyses indicated that *Thpok* was needed for the acquisition of the Tcmp cell signature at 7 dpi (Figures 2B and 2C), we examined whether *Thpok* supported the maintenance of CD4⁺ T memory cells or preserved memory potential at the initial phase of the response. To this end, we deleted *Thpok* either before or 3 weeks after LCMV Arm infection by using a tamoxifen-inducible *cre* transgene driven by *Cd4* cis-regulatory elements (*Cd4-creER*^{T2}). To eliminate the possibility of host-specific or indirect effects, we used chimeric mice containing *Zbtb7b*^{fl/fl} and control *Zbtb7b*^{+/+} components, both carrying the *Cd4-creER*^{T2} transgene. While loss of *Thpok* before infection impaired memory cell development, loss of *Thpok* after viral clearance had no detectable impact (Figures 4G and S4). Thus, *Thpok* regulates an early checkpoint in CD4⁺ T cell responders, protecting their memory potential in line with *Thpok*'s impact on the Tcmp cell signature.

Thpok Restrains a Dysfunction, Exhaustion-Like Circuitry

The RNA-seq analyses at 30 dpi showed that, in addition to expressing genes of the CD8⁺ T cell lineage and failing to establish

a memory program, *Thpok*-deficient CD4⁺ T cells expressed genes (including *Lag3*, *Havcr2*, and *Cd244* [respectively encoding *Lag3*, *Tim3*, and *2B4*; Figure 5A]) characteristic of dysfunctional T cells responding to chronic infection (Crawford et al., 2014; Wherry and Kurachi, 2015). Because of this mixed gene expression pattern, we explored the population heterogeneity at 30 dpi by performing scRNA-seq. t-SNE visualization of *Zbtb7b*^{AD} and control spleen GP66-specific T cells showed limited overlap between the two genotypes (Figure 5B). Unsupervised clustering identified four groups of control cells at 30 dpi; cells in three of these groups (clusters Ctrl1–3) expressed markers, including *Id3*, *Tcf7*, *Irf7*, and *Bcl2*, that are typical of memory cells. The fourth cluster (Ctrl4) showed memory-Tfh-cell attributes, including expression of *Bcl6* but not *Ccr7* (Figure 5C). None of the five *Thpok*-deficient clusters identified at 30 dpi (*Zbtb7b*^{AD} 1–5) expressed these memory markers or had a Tfh cell memory profile similar to cluster Ctrl4. These findings indicated that *Thpok* is important for the development of memory CD4⁺ T cells with Tfh and Th1 cell skewing. Three *Thpok*-deficient clusters (30 dpi *Zbtb7b*^{AD} 1–3) expressed effector genes, including *Prdm1* and *Tbx21*. Cluster *Zbtb7b*^{AD} 5, which included 18% of *Thpok*-deficient cells, additionally showed expression of *Havcr2*, *Lag3*, and *Pdcd1* (encoding PD-1) dysfunction genes. The remaining *Thpok*-deficient cluster, *Zbtb7b*^{AD} 4, associated Tfh cell (*Bcl6*), memory (*Sell*), and dysfunction (*Eomes* and *Lag3*) gene expression. Flow cytometry of *Thpok*-deficient cells at 30 dpi did not reproducibly detect expression of PD-1, an effect in line with PD-1's low expression by population RNA-seq (data not shown); however, the flow cytometry confirmed higher expression of *Tim3* and *2B4* relative to control cells (Figure 5D). Importantly, at 30 dpi, *Thpok*-deficient cells showed impaired IL-2 production (Figure 5E), a hallmark of T cell dysfunction (Crawford et al., 2014; Wherry and Kurachi, 2015). This dysfunctional profile was maintained during recall responses; *Thpok*-deficient LCMV-specific CD4⁺ T cells showed reduced production of IL-2 and increased expression of *Tim3* and *2B4* after clone 13 re-challenge (Figures S5A and S5B). We conclude from these analyses that *Thpok* restrains a dysfunction program resembling, but distinct from, that of exhausted T cells in chronic infections.

To examine whether *Thpok* targets the dysfunction program during the initial phase of the immune response, we defined a dysfunction signature on *Thpok*-deficient (*Zbtb7b*^{AD}) cells analyzed by scRNA-seq at 30 dpi (Table S2) and used this signature to query scRNA-seq results from cells at 7 dpi. We found higher scores for this signature in *Thpok*-deficient cells than in control cells (Figure S5C). Accordingly, adoptive-transfer experiments showed cell-intrinsic increases in the expression of *Tim3* and *2B4* and reduction in the production of IL-2 by

(C) Heatmap of differential gene expression (as defined in Figure S3F).

(D and E) Expression (fragments per million, fpm) of selected genes in indicated populations. p values are from a one-way ANOVA. (B, D, and E) Each symbol represents a biological replicate.

(F) Contour plots (left) and quantification (right, $n = 5$ per group) of CCR7 and CXCR5 expression on GP66:I-A^b spleen T cells from *Zbtb7b*^{AD}:WT mixed bone-marrow chimeras analyzed at 30 dpi with LCMV Arm. Data are from one representative experiment out of two.

(G) (left) Mixed bone-marrow chimeras (*Cd4-creER*^{T2} *Zbtb7b*^{fl/fl}, *Cd4-creER*^{T2} *Zbtb7b*^{+/+}) were infected with LCMV Arm after (Δ Early) or 3 weeks before (Δ Late) tamoxifen treatment and analyzed at 100 dpi (right). Percentage of GP66:I-A^b spleen T cells (*Thpok*-deleted or *Zbtb7b*^{+/+} control within the same chimeras) that were KLRG1^{hi} IL-7R α ^{lo} (Δ Early, $n = 5$; Δ Late, $n = 3$; Ctrl, $n = 8$). Data are from one experiment and representative of two independent mixed bone-marrow experiments and three similar experiments in non-chimeric mice; p values are from an unpaired, one-sided Welch's t test. Please see also Figures S3 and S4. Error bars indicate standard deviation.

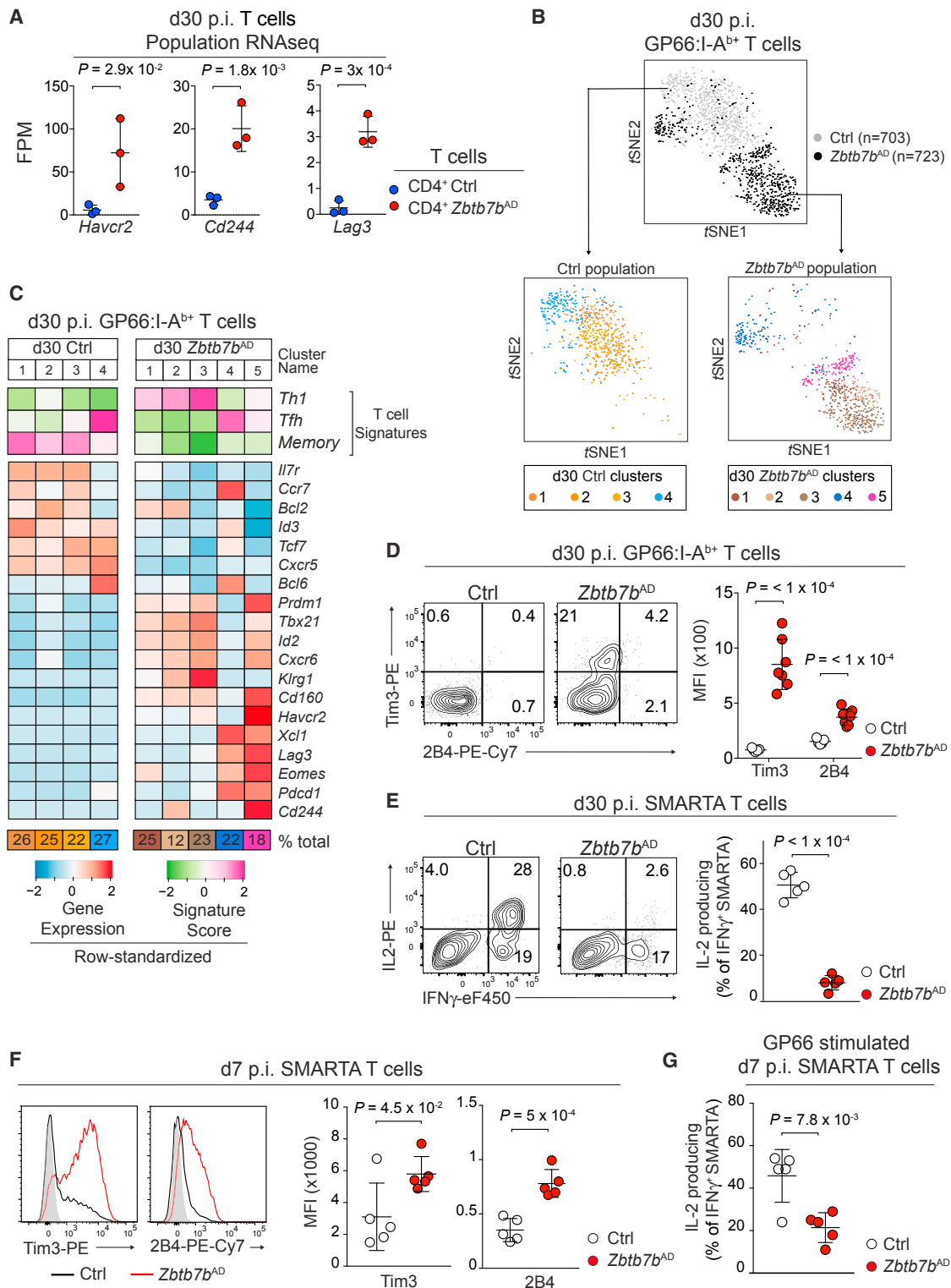


Figure 5. Thpok Is Needed for the Functional Fitness of CD4⁺ Memory T Cells

(A) RNA-seq expression of selected genes in *Zbtb7b*^{AD} or Ctrl GP66:I-A^b+ spleen T cells as in Figure 4D; p values are from a one-way ANOVA.

(B and C) scRNA-seq of *Zbtb7b*^{AD} or control (*Tnfrsf4-cre*⁺ *Zbtb7b*^{+/+}) spleen GP66:I-A^b+ *Rosa26*^{YFP} T cells at 30 dpi with LCMV Arm.

(B) t-SNE plots show all cells (top, colored by genotype) and each cell population after independent clustering (bottom, color code for clusters in [C, bottom]).

(C) Signature scores (top three rows) and expression of selected genes (bottom rows) among clusters of *Zbtb7b*^{AD} and control cells displayed in (B).

(legend continued on next page)

Thpok-deficient CD4⁺ T cells at 7 dpi (Figures 5F and 5G). Thus, during the initial response of CD4⁺ T cells to infection, Thpok both promotes the functional “fitness” of CD4⁺ T cells (by restraining a dysfunction transcriptional program) and preserves their memory potential.

Thpok Repression of *Prdm1* and *Runx3* Promotes Memory and Functional Fitness in CD4⁺ T Cells

To obtain mechanistic insight into Thpok control of the memory and dysfunction circuitries, we performed Thpok chromatin immunoprecipitation (ChIP) on effector CD4⁺ T cells. We generated, by CRISPR-Cas9 editing, a gene-targeted mouse carrying a *Zbtb7b*^{Bio} allele encoding a Thpok protein with a biotin acceptor site (Figure S6A). The *Zbtb7b*^{Bio} allele was functional, as it supported CD4⁺ T cell development in homozygous *Zbtb7b*^{Bio/Bio} mice (Figure S6B); furthermore, the encoded protein was efficiently biotinylated in CD4⁺ T cells that carried the *Zbtb7b*^{Bio} allele and expressed the BirA biotin ligase from the *Rosa26* locus (Figure S6C). Analyses of triplicate ChIP-seq samples obtained from *Zbtb7b*^{Bio} *Rosa26*^{BirA+} CD4⁺ T cells identified approximately 11,000 Thpok-bound genes in effector CD4⁺ T cells; most of these genes recruited Thpok at promoter regions (Figure S6D). Consistent with a previous report (Muroi et al., 2008), Thpok molecules bound the *Zbtb7b* and *Cd4* loci (Figure S6E and data not shown).

ChIP-seq detected Thpok binding to most of the Thpok-regulated genes identified in LCMV-specific CD4⁺ T cells by scRNA-seq at 7 dpi (Figure S6F). These notably included *Prdm1* and *Id2*, encoding two transcription factors promoting T cell effector differentiation and repressing memory development (Crotty et al., 2010; Rutishauser et al., 2009; Yang et al., 2011) (Figure 6A). We focused on *Prdm1* as a potential functional target of Thpok because its product, Blimp1, acts upstream of *Id2* in T cells (Miyazaki et al., 2014; Rutishauser et al., 2009; Shaw et al., 2016). Analyses of *Zbtb7b*^{AD} mice carrying a *Prdm1*^{YFP} reporter allele confirmed that Thpok restrains Blimp1 expression in LCMV-specific CD4⁺ T cells (Figures 6B and 6C). To examine whether Thpok protects memory potential or prevents dysfunction through *Prdm1* repression, we assessed the differentiation of *Zbtb7b*^{fl/fl} *Prdm1*^{fl/fl} *Tnfrsf4-cre* (*Zbtb7b*^{AD} *Prdm1*^{AD}) CD4⁺ T cells. To identify cell-intrinsic effects, we conducted most analyses in mixed bone-marrow chimeras. Inactivating *Prdm1* in Thpok-deficient cells restored an IL-7Rα^{hi} KLRG1^{lo} profile typical of WT memory CD4⁺ T cells at 90 dpi, and it enhanced IL-2 production (Figures 6D and 6E). Mirroring these late effects on functional fitness, *Prdm1* disruption fully reverted the Tim3 upregulation observed at 7 dpi in Thpok-deficient cells (Figure S7A), and there was a similar but incomplete effect on 2B4. In contrast, *Prdm1* disruption

only modestly enhanced CCR7 expression at 7 dpi and failed to restore long-term memory populations (Figures S7B and S7C). These findings suggested that Thpok repression of Blimp1 expression contributes to maintaining the functional fitness of memory cells but that it does not mediate the impact of Thpok on the generation of memory precursors.

In addition to having increased expression of Blimp1, Thpok-deficient CD4⁺ T cells at 7 dpi had increased expression of *Runx3* (Figure S7D), a factor expressed in CD8⁺ and Th1 CD4⁺ T cells (Djuretic et al., 2009) and previously shown to be a functional target of Thpok (Carpenter and Bosselut, 2010; Egawa and Littman, 2008; Kojo et al., 2017; Muroi et al., 2008; Wang et al., 2008a). Consistent with direct transcriptional control, we found that Thpok bound the *Runx3* locus (Figure 7A). Moreover, we noted from published ChIP-seq datasets (Mackay et al., 2016; Shan et al., 2017) that several genes, including *Tcf7* and *Ccr7*, that are characteristic of memory cells could recruit Blimp1 and *Runx* molecules in addition to Thpok (Figure S7E). Consequently, we speculated that maintaining memory potential required the repression of both *Prdm1* and *Runx3* by Thpok. To assess this possibility, we generated *Zbtb7b*^{fl/fl} *Prdm1*^{fl/fl} *Cbfb*^{fl/fl} *Tnfrsf4-cre* (*Zbtb7b*^{AD} *Prdm1*^{AD} *Cbfb*^{AD}) animals, in which deletion of the obligatory *Runx* cofactor *Cbfb* overcomes potential overlap of functions between *Runx1–3* proteins. We could not evaluate long-term memory responses in these mice because *Runx* activity is needed for long-term survival of CD4⁺ T cells (Egawa et al., 2007), including of memory CD4⁺ T cells (Figure S7F). Thus, we assessed LCMV-specific CD4⁺ T cells at 7 dpi for markers of memory potential (CXCR5 and CCR7) and of dysfunction (Tim3 and 2B4). We performed these analyses in mixed bone-marrow chimeras to avoid indirect effects of gene disruption. Triple-deficient cells differentiated normally into CCR7^{hi} CD4⁺ T cells, unlike their *Zbtb7b*^{AD} or *Zbtb7b*^{AD} *Prdm1*^{AD} counterparts (Figures 7B and 7C), whereas a partial rescue was observed in *Zbtb7b*^{AD} *Cbfb*^{AD} cells. Moreover, inactivation of both Blimp1 and *Cbfb* fully reverted the upregulation seen for inhibitory receptors Tim3 and 2B4 in Thpok-deficient cells (Figure 7D). We conclude from these experiments that Thpok promotes the emergence and functional fitness of memory CD4⁺ T cell precursors by restraining the expression or activity of Blimp1 and *Runx* molecules.

Altogether, our findings demonstrate that the transcription factor Thpok antagonizes Blimp1 and *Runx3* expression and thereby support CD4⁺ T cell memory potential and restrains a gene expression pattern characteristic of dysfunctional T cells. These functions of Thpok identify a shared mechanistic basis for two previously unconnected attributes of CD4⁺ helper T cells, ensuring their functional fitness to support long-lived immunity.

(D) Contour plots (left) and MFI (right) of Tim3 and 2B4 expression in GP66:1-A^b spleen T cells from *Zbtb7b*^{AD} (n = 7) and control (n = 5) mice at 30 dpi with LCMV Arm. Data are from two experiments and representative of three independent experiments.

(E–G) Naive *Zbtb7b*^{AD} and control SMARTA TCR transgenic cells were adoptively transferred into WT recipient one day before LCMV Arm infection.

(E) Cytokine production by spleen T cells at 30 dpi after peptide stimulation (left). The percentage of IL-2-producers among IFNγ⁺ cells is on the right (n = 5 recipient mice per genotype). Please see also Figure S5.

(F) Overlaid histograms (left) and MFI (right) of Tim3 and 2B4 expression in *Zbtb7b*^{AD} (red line) or control (black line) SMARTA spleen T cells at 7 dpi with LCMV Arm; gray-filled histograms show staining on CD44^{lo} CD4⁺ T cells from the same mouse (n = 5 per group).

(G) Percentage of IL-2 producers among IFNγ⁺ *Zbtb7b*^{AD} and control SMARTA spleen T cells after GP66 peptide stimulation at 7 dpi with LCMV Arm (n = 5 per genotype). (E–G) Data are gated on donor-derived cells and are from one experiment representative of two independent experiments per time point. Error bars indicate standard deviation.

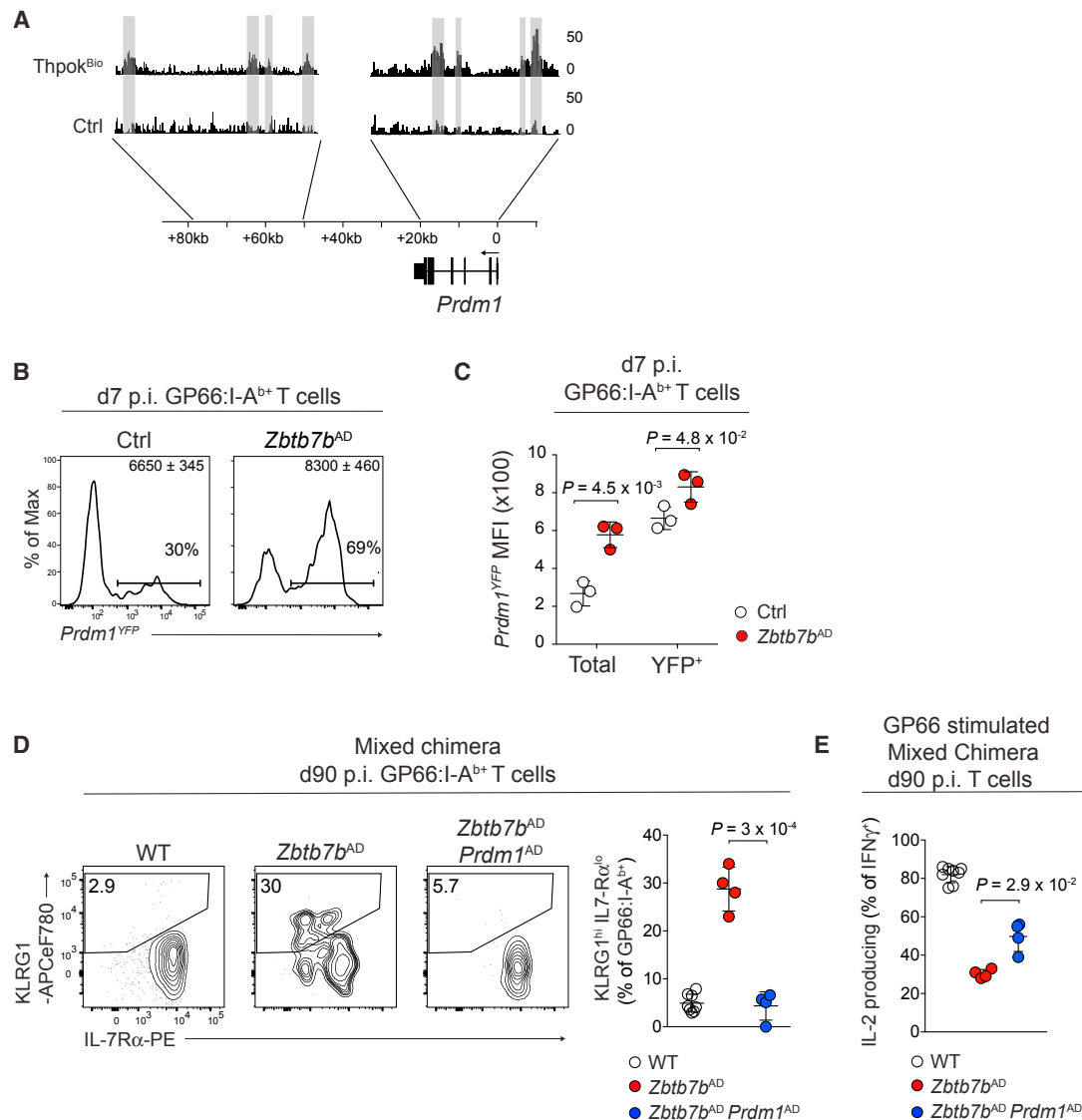


Figure 6. Thpok Maintains Memory CD4⁺ T Cell Fitness by Restraining Blimp1 Expression

(A) Thpok ChIP-seq in activated CD4⁺ T cells from *Zbtb7b*^{Bio} *Rosa26*^{BirA+} (*Zbtb7b*^{Bio}) and *Zbtb7b*^{+/+} *Rosa26*^{BirA+} (Ctrl) mice. Grey shading highlights peaks detected in three independent replicates.

(B) Histogram plots of yellow fluorescent protein (YFP) expression in GP66:I-A^{b+} spleen T cells from *Zbtb7b*^{AD} and control animals carrying the *Prdm1*^{YFP} reporter at 7 dpi with LCMV Arm.

Quantification (C) shows YFP expression in all (total) or in only YFP⁺ GP66:I-A^{b+} T cells. Data are from one experiment (n = 3 mice per genotype) representative of two independent experiments.

(D and E) *Zbtb7b*^{AD}:WT and *Zbtb7b*^{AD} *Prdm1*^{AD}:WT mixed bone-marrow chimeras were analyzed at 90 dpi with LCMV Arm.

(D) Contour plots (left) and quantification (right) of IL-7Rα versus KLRG1 surface expression in GP66:I-A^{b+} spleen T cells.

(E) Percentage of IL-2-producing cells among IFNγ⁺ spleen T cells of *Zbtb7b*^{AD} (n = 4), *Zbtb7b*^{AD} *Prdm1*^{AD} (n = 4), and WT (n = 8) genotype after GP66 peptide stimulation. Data are from one bone-marrow chimera experiment and representative of three independent non-chimeric experiments. Please see also [Figures S6](#) and [S7](#).

Error bars indicate standard deviation.

DISCUSSION

Memory CD4⁺ T cells are essential for durable immunity and are targeted by current vaccination strategies, including against influenza and malaria ([Mordmüller et al., 2017](#); [Valkenburg et al., 2018](#)). Thus, it is important to elucidate the mechanisms that control their development. Here, starting from single-cell

transcriptome analyses of virus-specific CD4⁺ T cells, we identified a regulatory switch controlling both the emergence of memory cells and their functional fitness. Generating transcriptome-wide gene expression signatures from large numbers of cells captured during the initial and memory phases of the immune response, we have provided a high-resolution analysis of effector and memory CD4⁺ T cell differentiation that had not

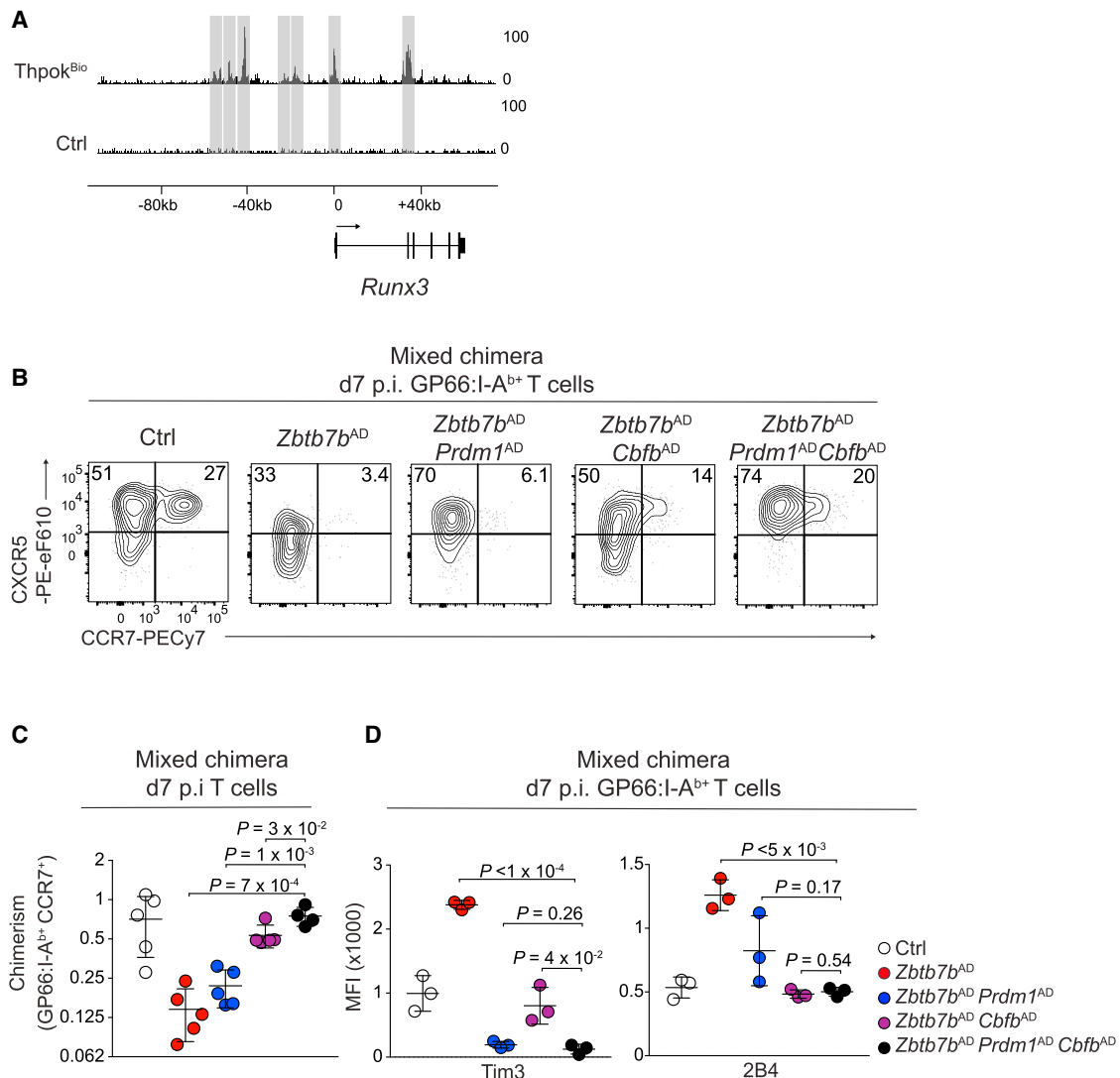


Figure 7. Memory CD4⁺ T Cell Differentiation Involves Thpok Antagonism of Blimp1 and Runx3

(A) Thpok ChIP-seq on the *Runx3* locus, displayed as in Figure 6A.

(B–D) Bone-marrow chimeras combining WT competitor and either *Zbtb7b*^{AD}, *Zbtb7b*^{AD} *Prdm1*^{AD}, *Zbtb7b*^{AD} *Cbfb*^{AD}, *Zbtb7b*^{AD} *Prdm1*^{AD} *Cbfb*^{AD}, or Ctrl “tester” cells were analyzed at 7 dpi with LCMV Arm.

(B) CCR7 and CXCR5 expression on GP66:I-A^{b+} tester spleen T cells.

(C) Tester:competitor ratios among spleen GP66:I-A^{b+} CCR7⁺ T cells (n = 5 per group).

(D) Tim3 and 2B4 expression on GP66:I-A^{b+} cells (n = 3 per group). (C and D) Data are representative of one (D) or two separate (C and D) experiments on one set of mixed chimeras and three similar experiments on non-chimeric animals of the same genotype. Please see also Figures S6 and S7.

Error bars indicate standard deviation.

been accessible to previous single-cell (Lönnberg et al., 2017) and population-level studies (Hale et al., 2013; Marshall et al., 2011).

At the initial phase of the response, we identified characteristic Th1 and Tfh cell transcriptomic signatures, which included genes encoding prototypical fate markers (e.g., T-bet and Id2 for Th1 cells or CXCR5 and Icos for Tfh cells); these signatures were largely mutually exclusive, consistent with the idea that Th1- and Tfh-effector-cell fates show little functional overlap (Crotty, 2018). The Tcmp cell signature was associated with memory cell markers (including *Ccr7*, *Tcf7*, or *Bcl2*) and excluded genes

(*Tbx21*, *Prdm1*, *Bcl6*) that are characteristic of Th1- and Tfh-effector-cell genes, supporting its assignment to cells with increased memory potential. Accordingly, that signature mapped to a subset of CD4⁺ T cells expressing CCR7 (and also CXCR5, although in intermediate amounts) and were previously reported to contain memory precursor activity (Pepper et al., 2011). Partial activation of the Tcmp cell signature was detected in Th1 but not Tfh cell subsets, suggesting that it identifies cells skewed toward a Th1- rather than Tfh-memory-cell potential. Consistent with this idea, two Tcmp-signature components, *Klf2* and *Ccr7*, inhibit Tfh cell differentiation (Haynes et al.,

2007; Lee et al., 2015). The possibility of an early divergence of central and Tfh-memory-cell precursors does not exclude the fact that their differentiation programs share transcriptional regulators. Bcl6, which is needed for the generation of memory Tfh cells (He et al., 2013), promotes the survival of all memory CD4⁺ T cell subsets (Ichii et al., 2007). Additionally, Bcl6 is needed for the generation of the CXCR5⁺ subset during the initial CD4⁺ T cell response (Pepper et al., 2011); although this might simply reflect the importance of Bcl6 for CXCR5 expression, it could also indicate its involvement in the emergence of CD4⁺ T cell memory, in addition to its involvement in long-term memory cell survival.

In addition to its cell-intrinsic effect on CD4⁺ T cell memory, Thpok is needed for CD4⁺ T cell help for CD8⁺ T cell memory, a function critical for long-term immunity (Shedlock and Shen, 2003; Sun and Bevan, 2003). To account for CD4⁺ T cell help for CD8⁺ T cells, others have proposed several non-mutually exclusive mechanisms, including IL-2 production and the “licensing” of dendritic cells, through CD40 signaling, for the provision of memory-inducing signals to CD8⁺ T cells (Laidlaw et al., 2016; MacLeod et al., 2009). Both IL-2 secretion and expression of CD154, the ligand for CD40, depend on Thpok, accounting at least in part for its contribution to CD4⁺ T cell help with the generation of memory CD8⁺ T cells.

Thpok protects the functional fitness of CD4⁺ memory cells by restraining a ‘dysfunction’ gene-expression program that shares features with the ‘exhausted’ state of T cells subject to chronic antigen stimulation; these features include impaired IL-2 production and dependence on Blimp1 (Crawford et al., 2014; Shin et al., 2009; Wherry and Kurachi, 2015). However, this dysfunction pattern differs from conventional exhaustion in that it emerges despite antigen clearance, e.g., in mixed bone marrow chimeras or adoptively transferred cells; accordingly, it is not accompanied by PD1 expression, a hallmark of recent antigen engagement (Agata et al., 1996). Additionally, the Thpok-repressed dysfunction pattern is distinct from the cytotoxic diversion we and others previously identified in Thpok-deficient CD4⁺ T cells (Mucida et al., 2013; Reis et al., 2013; Vacchio et al., 2014; Wang et al., 2008a), most notably because the former depends on Blimp1, whereas the latter depends on Runx activity.

Repression of Blimp1- and Runx3-encoding genes mediated the impact of Thpok on the emergence and functional fitness of memory CD4⁺ T cells. Our observation that Thpok binds within or near both loci supports the possibility that their repression by Thpok is, at least in part, direct. Thpok repression of Blimp1 feeds into a negative regulatory loop whereby Bcl6 and Blimp1 inhibit each other’s expression (Crotty et al., 2010), previously characterized in differentiating CD4⁺ effectors. In addition, we found that Thpok binds the *Bcl6* gene (unpublished data), whereas Runx3 was reported to shift the Bcl6-Blimp1 equilibrium toward Blimp1 expression (Shan et al., 2017) so that its repression could reinforce direct effects of Thpok on the Bcl6-Blimp1 loop. Together, these observations support the idea that Thpok serves at multiple entry points to shift the Bcl6-Blimp1 loop to restrain Blimp1.

In summary, the present study identified a Thpok-operated checkpoint controlling both the emergence of CD4⁺ memory cells and their functional fitness, two previously unconnected aspects of CD4⁺ T cells responses. Although Thpok is expressed in

all conventional CD4⁺ T cells, it is sensitive to both antigenic stimulation (He et al., 2008; Wang et al., 2008a) and environmental factors, as was recently shown of gut microbial metabolites (Cervantes-Barragan et al., 2017; Reis et al., 2013). Thus, the importance of Thpok in CD4⁺ T cell memory and functions identifies this factor as a potential research target that could enhance responses to immunization strategies against infections or cancer.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and two tables and can be found with this article online at <https://doi.org/10.1016/j.immuni.2018.12.019>.

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AUTHOR CONTRIBUTIONS

T.C. and R.B. designed research; T.C., M.S.V., and Y.G. performed research and analyzed data. M.P. performed initial experiments. F.T.M. and L.T. generated the *Zbtb7b*^{Bio} allele. M.M., Y. Z., and B.T. contributed to initial scRNA-seq capture and analyses. T.C. performed bioinformatics analyses with guidance from J.C. D.B.M. provided viral stocks and guidance with LCMV experiments. T.C. and R.B. wrote the manuscript with input from other co-authors. R.B. supervised the research.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-B220-V500 (Clone RA3-6B2)	BD Pharmigen	Cat# 561227, RRID:AB_10562193
Anti-CD45.1-eFluor450 (Clone A20)	Thermofisher	Cat# 48-0453-82, RRID:AB_1272189
Anti-CD45.2-PE-Cy7 (Clone 104)	BD Pharmigen	Cat# 560696, RRID:AB_1727494
Anti-CD5-PerCP-Cyanine5.5 (Clone 53-7.3)	Thermofisher	Cat# 45-0051-82, RRID:AB_914334
Anti-CD4-PE-Cy7 (Clone GK1.5)	Thermofisher	Cat# 25-0041-82, RRID:AB_469576
Anti-CD8 α -eFluor450 (Clone 53-6.7)	Thermofisher	Cat# 48-0081-82, RRID:AB_1272198
Anti-KLRG1-APC-eFluor780 (Clone 2F1)	Thermofisher	Cat# 47-5893-82, RRID:AB_2573988
Anti-CCR7-PE-Cy7 (Clone 4B12)	Thermofisher	Cat# 25-1971-82, RRID:AB_469652
Anti-CXCR5-PE-eFluor 610 (Clone SPRCL5)	Thermofisher	Cat# 61-7185-82, RRID:AB_2574660
Anti-IL7-R α -PE (Clone A7R34)	Thermofisher	Cat# 12-1271-83, RRID:AB_465845
Anti-IL7R α -APC (Clone A7R34)	Thermofisher	Cat# 17-1271-82, RRID:AB_469435
Anti-Foxp3-AF488 (Clone FJK-16 s)	Thermofisher	Cat# 53-5773-82, RRID:AB_763537
Anti-IL2-PE (Clone JES6-5H4)	Thermofisher	Cat# 12-7021-82, RRID:AB_466150
Anti-IFN γ -eFluor450 (Clone XMG1.2)	Thermofisher	Cat# 48-7311-82, RRID:AB_1834366
Anti-CD44-APC-eF780 (Clone IM7)	Thermofisher	Cat# 47-0441-82, RRID:AB_1272244
Anti-Tim3-PE (Clone RMT3-23)	Biolegend	Cat# 119704, RRID:AB_345378
Anti-2B4-PE-Cy7 (Clone ebioR244F4)	Thermofisher	Cat# 25-2441-80, RRID:AB_2573431
Anti-CXCR6-APC (Clone SA051D1)	Biolegend	Cat# 151106, RRID:AB_2572143
Anti-GranzymeB-PE (Clone GB12)	Thermofisher	Cat# 25-2441-80, RRID:AB_2573431
Streptavidin	Thermofisher	Cat#12-4317-87
Anti-Thpok (Clone T43-94)	BD Pharmigen	Cat#565500
Bacterial and Virus Strains		
LCMV Armstrong	McGavern Lab	Generated in house
LCMV Clone 13	McGavern Lab	Generated in house
Chemicals, Peptides, and Recombinant Proteins		
I-A ^b LCMV GP66-77 (DIYKGVYQKSV) tetramer	NIH tetramer facility	n/a
H-2D ^b LCMV GP33-42 (KAVYNFATM) tetramer	NIH tetramer facility	n/a
GP33 peptide (KAVYNFATM)	Anaspec	AS-61296
GP66 peptide (GLKGPDYKGVYQFKSVEFD)	Anaspec	AS-64851
Formaldehyde	Thermofisher	28906
Critical Commercial Assays		
Transcription factor Staining Buffer Set	Thermofisher	00-5523-00
Golgi Stop	BD Pharmigen	554724
Protein-A beads	Invitrogen	10001D
Streptavidin beads (M280)	Invitrogen	12205D
Micro RNeasy columns	QIAGEN	74034
Chromium single cell 3' v2 kit	10X Genomics	120267, 120262
Proteinase K	Invitrogen	25530049
RNase A	Invitrogen	1209121
Deposited Data		
RNaseq	This study	GEO:GSE116519
CHIPseq (Thpok)	This study	GEO:GSE116506
Single-cell RNaseq	This study	GEO:GSE121002

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
CHIPseq (CBFb)	Shan et al. (2017)	GEO:GSE81888
CHIPseq (Blimp1)	Mackay et al. (2016)	GEO:GSE79339
Experimental Models: Organisms/Strains		
<i>Zbtb7b</i> ^{fl/fl}		Wang et al. (2008a)
<i>Zbtb7b</i> ^{-/-}		Wang et al. (2008b)
<i>Zbtb7b</i> ^{Bio}	This study	N/A
<i>Rosa26</i> ^{BirA}		Driegen et al. (2005)
<i>Prdm1</i> ^{fl/fl}		Shapiro-Shelef et al. (2003)
<i>Cbfb</i> ^{fl/fl}		Naoe et al. (2007)
<i>Cd4sil</i> ^{fl/fl}		Zou et al. (2001)
<i>Tnfrsf4-cre</i>		Zhu et al. (2004)
<i>Cd4-creER</i> ^{T2}		Aghajani et al. (2012)
SMARTA		Oxenius et al. (1998)
<i>Prdm1</i> ^{YFP}	Jackson Laboratory	Jax 008828
<i>Rosa26</i> ^{YFP}		Srinivas et al. (2001)
NCI B6-Ly5.1/Cr (CD45.1)	Charles River	Charles River 564
C57BL/6Ncr (CD45.2)	Charles River	Charles River 556
B6 CD45.1 CD45.2		Generated in house
Software and Algorithms		
Graphpad Prism 7.0	graphpad.com	N/A
Flowjo 10.0	flowjo.com	N/A
R	r-project.org	N/A
Seurat v 2.3.0		Butler et al. (2018)
MACS2		github.com/taoliu/MACS
CellRanger v1.3.1	10X Genomics	N/A
DESeq2		github.com/mikelove/DESeq2
PartekFlow	partek.com	N/A
STAR aligner		github.com/alexdobin/STAR
ChIPseeker		github.com/GuangchuangYu/ChIPseeker

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Rémy Bosselut (remy.bosselut@nih.gov).

METHOD DETAILS**Mice**

Mice carrying floxed alleles for *Zbtb7b* (Wang et al., 2008a), *Cbfb* (Naoe et al., 2007), *Prdm1* (Shapiro-Shelef et al., 2003), *Cd4sil* (Zou et al., 2001), *Rosa26*^{YFP} (Srinivas et al., 2001), or *Rosa26*^{BirA} (Driegen et al., 2005) were previously described as were *Zbtb7b*^{-/-} (Wang et al., 2008b), *H2-Ab1*^{b-/-} (Grusby et al., 1991), SMARTA (Oxenius et al., 1998), *Cd4-creER*^{T2} (Aghajani et al., 2012) and *Tnfrsf4-cre* (Zhu et al., 2004) animals. *Prdm1*^{YFP} mice were from the Jackson laboratory and CD45.1 and CD45.2 C57BL/6 mice from Charles River laboratories. All transgenic mice were heterozygous for the transgene they express. The *Tnfrsf4-cre* allele was maintained heterozygous and only female *Tnfrsf4-cre*⁺ mice were used for breeding. Except where specified otherwise, control mice included in experimental designs were *Tnfrsf4-cre*⁺ *Zbtb7b*^{+/+} or *Tnfrsf4-cre*⁻ animals from the same line as experimental mice. Mice were housed in specific pathogen-free facilities. Animal procedures were approved by the NCI Animal Care and Use Committee.

Infectious Models, Mixed Bone Marrow Chimera, and Induced Gene Deletion

Non-chimeric 6-16 weeks old animals were infected by intra-peritoneal injection of 2×10^5 pfu of LCMV Armstrong. For re-challenge experiments mice were infected intra-venously with 2×10^6 pfu of LCMV Clone 13.

For adoptive transfer experiments, 1000 spleen CD4⁺ T cells (unseparated or sorted CD44^{lo} Rosa26^{YFP}) from SMARTA TCR transgenic mice (*Zbtb7b*^{AD} or controls) were intra-venously injected into CD45 congenic recipients, which were infected with LCMV Arm 24 hours later.

For mixed bone marrow chimera experiments, T cell-depleted (Pan T Dynal kit, Invitrogen) bone marrow cells were prepared from CD45 disparate mice, mixed together at various ratios (from 1:1 to 1:3), and injected into lethally irradiated (900 rads) CD45.1 recipients. Chimeric animals were infected eight weeks after reconstitution. In experiments using *Cd4*^{ERT2Cre} animals, gene disruption was induced by tamoxifen gavage (5mg/mouse/day for 4 consecutive days) before or 3 weeks after LCMV Arm infection. Chimerism was calculated as the ratio of tester over competitor cells within the population of interest.

Determination of Virus Titers

Serum from infected animals was isolated from peripheral blood by centrifugation and stored at -80°C . Viral RNA was extracted with PureLink Viral RNA column (Invitrogen) according to the manufacturer instructions, reverse-transcribed (Thermoscript, Invitrogen), and quantified by real-time PCR with SYBR green dye (Invitrogen) on a QuantStudio 6 Flex system (Applied Biosystems), using primers specific for the LCMV GP gene (McCausland and Crotty, 2008) (Fwd-GCAACTGCTGTGTTCCCGAAAC, Rev-CATT CACCTGGACTTTGTCAGACTC). For each experiment, the limit of detection (LOD) of the assay was determined using the serum of non-infected animals.

Antibodies

Antibodies for the following specificities were purchased either from Becton Dickinson PharMingen, Biolegend or ThermoFisher Ebiosciences: CD4 (RM4.4 or GK1.5), CD8 α (53-6-7), CD45.2 (104), CD45.1 (A20), TCR β (H57-597), CD5 (53-7.3), B220 (RA3-6B2), IFN γ (XMG1.2), IL-2 (JES6-5H4), CD44 (IM7), IL-7R α (A7R34), KLRG1 (2F1), CCR7 (4B12), CXCR5 (SPRCL5), Tim3 (RMT3-23), 2B4 (R244F4), CXCR6 (SA051D1), streptavidin. MHC tetramers loaded with the LCMV GP33 or GP66 peptides were obtained from the NIH Tetramer Core Facility.

Cell Preparation and Staining

Spleen cells were prepared and stained as previously described (Ciucci et al., 2017). Surface staining with GP66:I-A^b tetramer or for CCR7 or CXCR5 was performed at 37°C , and staining with GP33:H-2D^b tetramer at 4°C , for 1 hour prior to staining with other antibodies. Cytokine staining was performed after 5 hours on splenocytes incubated in the presence of GP66 (GLKGPDIYKGVYQFKS VEFD, 2 $\mu\text{g/mL}$) or GP33 (KAVYNFATM, 0.2 $\mu\text{g/mL}$) peptides (Anaspec) and Golgi Stop (Crawford et al., 2014).

Flow cytometry data was acquired on LSR II or LSR Fortessa cytometers (BD Biosciences) and analyzed with FlowJo software (TreeStar). Dead cells and doublets were excluded by LiveDead staining (Invitrogen) and forward scatter height by width gating. Except if otherwise mentioned, numbers in the cytometry plot are percentages of cells in the gates. Cell sorting was performed on a FACS Aria or FACS Fusion (BD Biosciences). Because of the effect of Thpok-deletion on the expression of CD4 and CD8 α , analyses and purification of GP66-specific cells were performed without gating for CD4 and CD8 α expression.

Population RNA-Seq

RNA from GP66:I-A^{b+} and GP33:H-2D^{b+} T cells, sorted from the spleen of LCMV-infected mice at d30 p.i., was isolated using the RNeasy Plus Micro kit (QIAGEN). Quality control was performed by bioanalyzer (Agilent), and RNA samples with a RNA integrity number (RIN) > 8 were processed for library preparation using SMARTer Ultra Low Input reagent (Takara) and Nextera XT DNA (Illumina) library preparation kits. Libraries were sequenced with paired-end reads of 126bp on a HiSeq2500 sequencer (Illumina) to reach 50 million read pairs per sample. For each cell subset and genotype, data are derived from three distinct mice, with separate processing from cell sorting to sequencing. Raw RNaseq fastq reads were trimmed with Trimmomatic and aligned to mouse genome (mm10) using STAR (v. 2.4.0h). Gene-assignment and count of RNA reads were performed with HTseq. Further analyses were performed with R software and differentially expressed genes were identified using DESeq2 using the Wald test (FDR < 0.01).

Generation of a *Zbtb7b*^{Bio} Genetically Targeted Allele

Specific sgRNAs targeting the 3' end of the *Zbtb7b* open reading frame were designed using the online tool MIT CRISPR Design (crispr.mit.edu) and produced *in vitro* using MEGAshortscript T7 transcription Kit (Thermo Fisher Scientific). A 153bp double stranded template containing the composite linker-biotin acceptor sequence agcatgctcggtggaGGCCTGAACGACATCTTCGAGGCTCA GAAATCGAATGGCAGCAA (one-letter code amino acid sequence smrsggGLNDIFEAQKIEWHE, biotin acceptor peptide sequence in uppercase letters) flanked by 45bp left and right Thpok homology regions was synthesized as Ultramer DNA oligo (Integrated DNA Technologies). sgRNAs, dsDNA donor template and Cas9 mRNA (TriLink Biotechnologies) were microinjected into zygotes from C57BL/6Ncr mice at the one-cell stage to generate the *Zbtb7b*^{Bio} allele.

ChIP-Sequencing

Splenic CD4⁺ T cells from *Zbtb7b*^{Bio/+} Rosa26^{BirA} or Rosa26^{BirA} animals were enriched using Dynabeads Untouched Mouse CD4 cells kit (Invitrogen) and stimulated with anti-CD3 (1 $\mu\text{g/mL}$), anti-CD28 (3 $\mu\text{g/mL}$) and IL-12 (10ng/mL) for 3 days and then with IL-2 (100ng/mL) for another day. Live cells were sorted, washed in PBS and fixed in 1% formaldehyde-containing PBS for 5 min at 37 degrees. Following quenching with 0.125M glycine, cells pellets were snap-frozen on dry ice and stored at -80°C . Frozen fixed

cells were then lysed in 1% SDS RIPA buffer (20 mM Tris-HCl pH 7.6, 2 mM EDTA, 150 mM NaCl, 0.1% sodium deoxycholate, 1% Triton X-100) for 30 min at 4°C, spun down and sonicated in 0.1% SDS RIPA buffer using a Qsonica Q800R sonicator (30 s on, 59 s off, 85% amplitude) to obtain a sheared chromatin with an average size of 200bp. Sheared chromatin was pre-cleared with protein-A magnetic beads (Invitrogen 10001D) followed by immunoprecipitation with M280 Streptavidin beads (Invitrogen 11205D). After washing, immunoprecipitated chromatin was reverse-crosslinked by overnight incubation at 70°C in 50nM 50mM Tris-HCl pH 8., 10mM EDTA, 1% SDS buffer. DNA was subsequently treated with proteinase K and RNase (both at 0.2mg/mL) before purification using the QIAquick PCR purification kit (QIAGEN).

DNA was processed for library preparation using the Accel-NGS 2S DNA reagent (Swift). Libraries were sequenced (75bp single-ended reads) on a NextSeq sequencer (Illumina). Raw fastq reads were trimmed based on quality score (Phred > 32) and aligned to mouse genome (mm10) using Bowtie on Partek Flow on the National Institutes of Health high-performance computing Biowulf cluster. Peak calling was performed with MACS2 comparing the ChIP samples from the *Zbtb7b*^{Bio} *Rosa26*^{BirA} chromatin to the *Rosa26*^{BirA} chromatin (broad peak, qvalue < 0.1). Further analyses were performed with R software with the ChIPseeker package.

The Cbfb (Shan et al., 2017) and Blimp1 (Mackay et al., 2016) datasets were obtained from the gene expression omnibus (respectively accession GSE81888 and GSE79339) and aligned to the mouse genome (mm10 release) using the Bowtie package on the Partek Flow server.

Single-Cell RNA-seq

5000-10000 splenic T cells (either a mix of CD4⁺ and CD8⁺ T cells or GP66:I-A^{b+} T cells, as indicated in Figures) were sorted from LCMV infected mice, loaded onto a 10X Chromium platform to generate cDNAs carrying cell- and transcript-specific barcodes that were used to construct sequencing libraries using the Chromium Single Cell 3' Library & Gel Bead Kit v2 according to the manufacturer instructions. Libraries were sequenced on multiple runs of Illumina NextSeq using paired-end 26x98bp or 26x57bp in order to reach a sequencing saturation greater than 70%, resulting in at least 50,000 reads/cell. Single-cell sequencing files were processed, and count matrixes extracted using the Cell Ranger Single Cell Software Suite (v1.3.1). Further analyses were performed in R using the Seurat package (Butler et al., 2018) (2.3.0).

Data was pre-processed by removing genes expressed in fewer than 2 cells and excluding cells expressing fewer than 500 genes, or more than 10% mitochondrial genes. Reduction of data dimensionality was performed on the first 20 principal components (PC) calculated on the highly variable genes for each pre-processed dataset.

To define gene expression signatures, we first clustered d7 p.i CD44^{hi} datasets (including both CD8⁺ and CD4⁺ T cells) and defined as CD4⁺ clusters those clusters scoring high for *Cd4* and negative or low for *Cd8a*. Cells from these clusters were extracted from the whole set. Highly variable genes were re-defined on the selected cells and *Cd4*-expressing populations were clustered. For each cluster, a cluster-specific gene signature was defined using the *FindAllMarkers* function selecting only positively enriched genes. The final Th1, Tfh, Tcmp, Treg and memory cell signatures were defined as the intersection of matching cluster-specific gene signatures identified in the two independent replicates. Gene signature scores were calculated on regressed datasets (regression of the number of UMI and percentages of mitochondrial genes with *ScaleData* function) using the *AddModuleScore* function.

For analyses of Thpok-deficient and control cells, clustering was performed separately for each sample, and further analyses, including t-SNE visualization, were performed after merging datasets using the *MergeSeuratObjects* function. Clusters representing less than 1% of each population (which corresponds to the expected duplicate capture rate) were excluded from downstream analyses.

QUANTIFICATION AND STATISTICAL ANALYSIS

Except for deep-sequencing data, statistical significance was calculated with Prism software. Except where otherwise indicated in figure legends, error bars in graphs indicate standard deviation and statistical comparisons were done by unpaired two-sided Welch's t test.

DATA AND SOFTWARE AVAILABILITY

The accession numbers for the sequence data reported in this paper are : GEO:GSE116506, GSE116519, GSE121002 for Thpok ChIP-seq, RNA-seq, and single-cell RNA-seq, respectively. All other data and code are available upon request.

Immunity, Volume 50

Supplemental Information

The Emergence and Functional Fitness

of Memory CD4⁺ T Cells

Require the Transcription Factor Thpok

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Supplemental Figure 1, Related to Figure 1

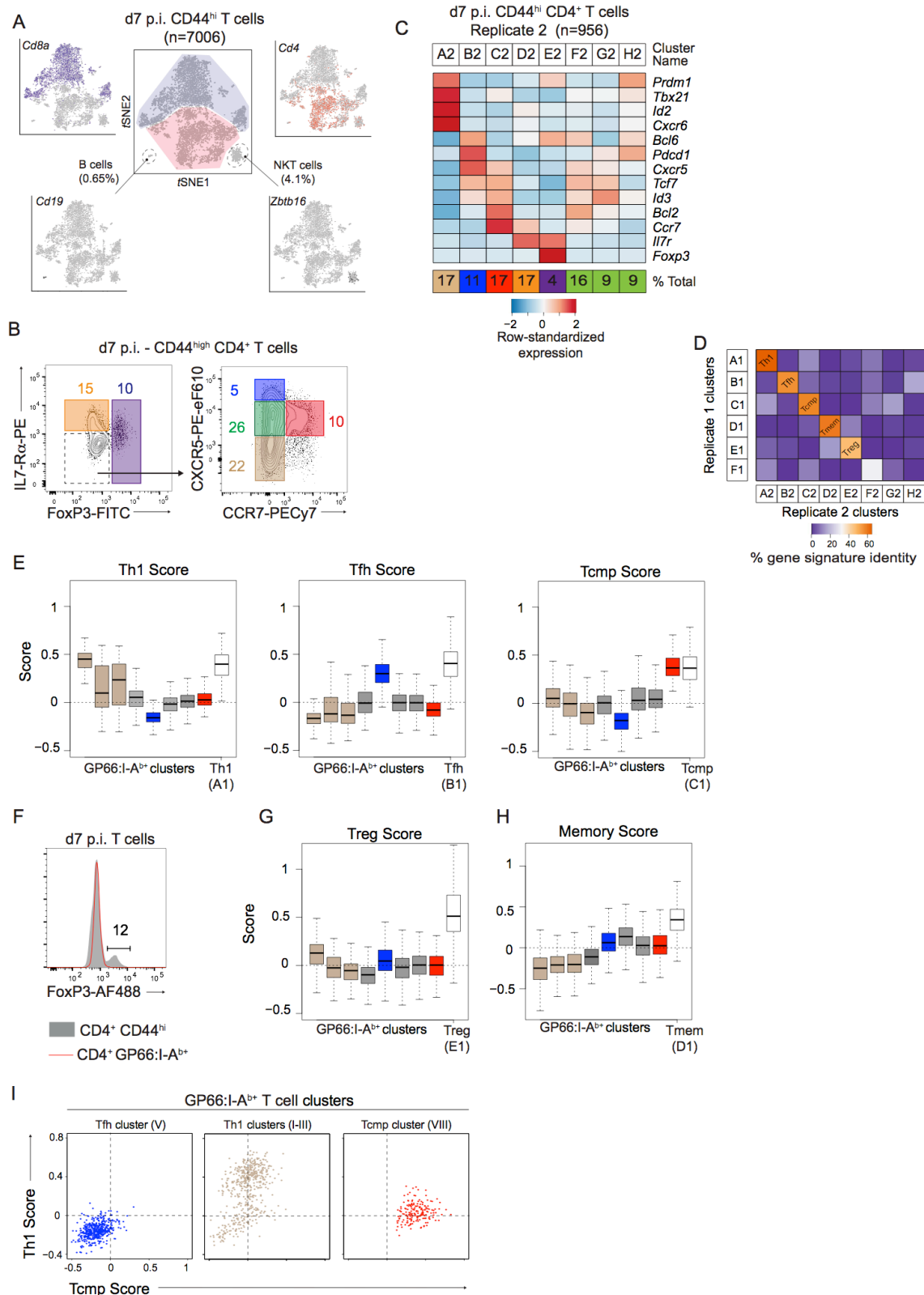


Figure S1: Identification and validation of CD4⁺ signatures by scRNAseq.

(A) CD44^{hi} spleen T cells from WT mice at d7 p.i. with LCMV Arm were analyzed by scRNAseq. Plot displays single CD44^{hi} T cells from experimental replicate 1, visualized with the t-SNE algorithm. Blue and red areas highlight the *Cd8a* and *Cd4* expressing cells respectively. Circles show B cells and iNKT cells, expressing *Cd19* and *Zbtb16* (encoding PLZF), respectively.

(B) Plots show the expression of CCR7 and CXCR5 on Foxp3⁻ IL-7R α ^{lo} spleen CD44^{hi} CD4⁺ T cells at d7 p.i. with LCMV Arm. Gates are colored according the code used for cell clusters defined in Fig. 1B, and numbers show the percentage of each subsets relative to the total population.

(C) Second replicate scRNAseq analysis of spleen WT CD44^{hi} T cells at d7 p.i. with LCMV Arm. Heatmap shows the relative expression of selected genes on the 8 unsupervised CD4⁺ clusters.

(D) Correlation matrix shows the percentages of identical genes between the gene signatures among the clusters of CD4⁺ T cells detected in Replicate 1 (Clusters A1 to F1, as shown in Fig. 1B, C) and in Replicate 2 (Clusters A2 to H2, Fig. S1C).

(E, G, H) Cells clusters from d7 p.i. GP66:I-A^b⁺ WT spleen T cells (defined in Fig. 1F, G) are scored for the cell signatures. Boxplots show the centered distribution of cell scores for each cell signature. Clusters with high scores for Th1- (Clusters I-III), Tfh- (Cluster V) and Tcmp- (Cluster VIII) cell signatures are colored in brown, blue and red respectively. As a reference, the rightmost white box in each plot indicates the score of cells from the corresponding, cell signature-defining cluster in Replicate 1 (Cluster A1-E1, Fig. 1B, C).

(F) Plot shows Foxp3 expression on spleen CD4⁺ CD44^{hi} and CD4⁺ GP66:I-A^b⁺ T cells from WT mice at d7 p.i. with LCMV Arm. The percentage of Foxp3⁺ among CD4⁺ CD44^{hi} T cells is shown.

(I) Dot plots show centered scores for Tcmp (x-axis) vs. Th1 (y-axis) signatures on cells from Cluster V (Tfh), Clusters I-III (Th1) and Cluster VIII (Tcmp) defined in Fig. 1F, G. Each dot represents a cell.

Supplemental Figure 2, Related to Figures 2 and 3

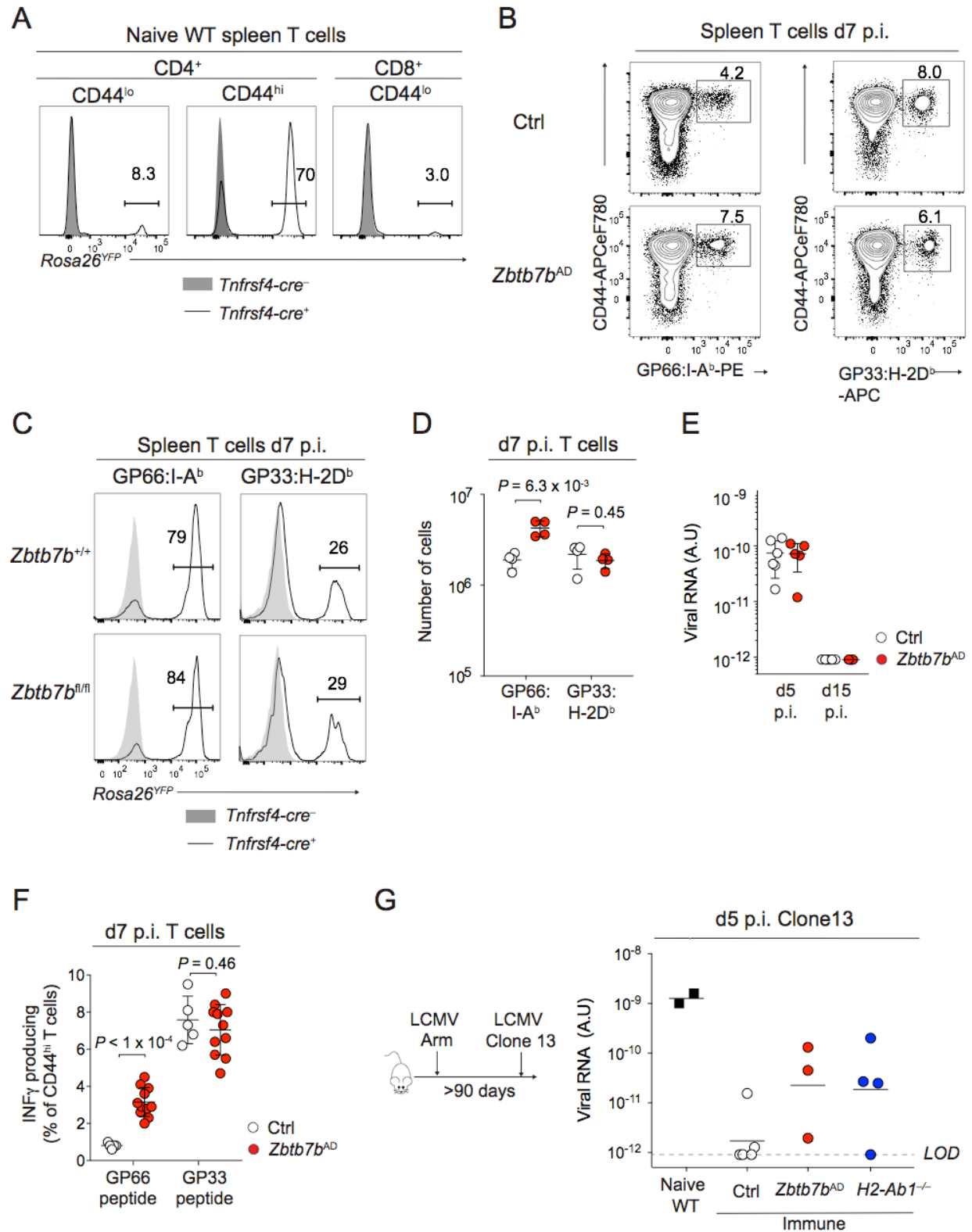


Figure S2: Post-activation disruption of *Zbtb7b* in LCMV-infected *Zbtb7b*^{AD} mice.

(A) Plots show Cre activity as reported by *Rosa26*^{YFP} expression on the indicated spleen T cell populations from *Tnfrsf4-cre*⁺ (black line) or *Tnfrsf4-cre*⁻ (grey histogram) uninfected animals. Data are representative of two independent experiments.

(B) Plots show GP66:I-A^b and GP33:H-2D^b staining at d7 p.i. with LCMV Arm on total spleen T cells from the indicated mice.

(C) Cre activity as reported by YFP expression was measured on spleen GP66:I-A^{b+} and GP33:H-2D^{b+} T cells from *Tnfrsf4-cre⁻ Rosa26^{YFP}* (grey histogram) or *Tnfrsf4-cre⁺ Rosa26^{YFP}* (black line) mice of the indicated genotype at d7 p.i.

(D) Graph shows the numbers of spleen GP66:I-A^{b+} and GP33:H-2D^{b+} T cells in *Zbtb7b^{AD}* (n=4) and control (n=5) mice analyzed as in (B). P-values from unpaired two-sided Welch's *t*-test are indicated. Data (B-D) are representative of two independent experiments.

(E) Serum virus titer at d5 and d15 p.i. of *Zbtb7b^{AD}* (n=5) and control (n=6) mice with LCMV Arm. Data are representative of two independent experiments.

(F) Percentage of IFN γ ⁺ cells among *Zbtb7b^{AD}* (n=11) and control (n=5) CD44^{hi} spleen T cells after GP66 (MHC-II restricted) and GP33 (MHC-I restricted) peptide stimulation. P-values from unpaired two-sided Welch's *t*-test are indicated. Data are representative of two independent experiments.

(G) *Zbtb7b^{AD}*, *H2-Ab1^{-/-}* (MHC-II-deficient) or control mice were infected with LCMV Arm and re-challenged with LCMV Clone 13 more than 90 days later. Graph shows serum virus titers in animals of the indicated genotype (red, blue and empty circles) 5 days after Clone 13 infection; black squares show titers in naive WT mice as controls. Dashed line shows the limit of detection (LOD) of the assay. Data are representative of two independent experiments.

Supplemental Figure 3, Related to Figure 4

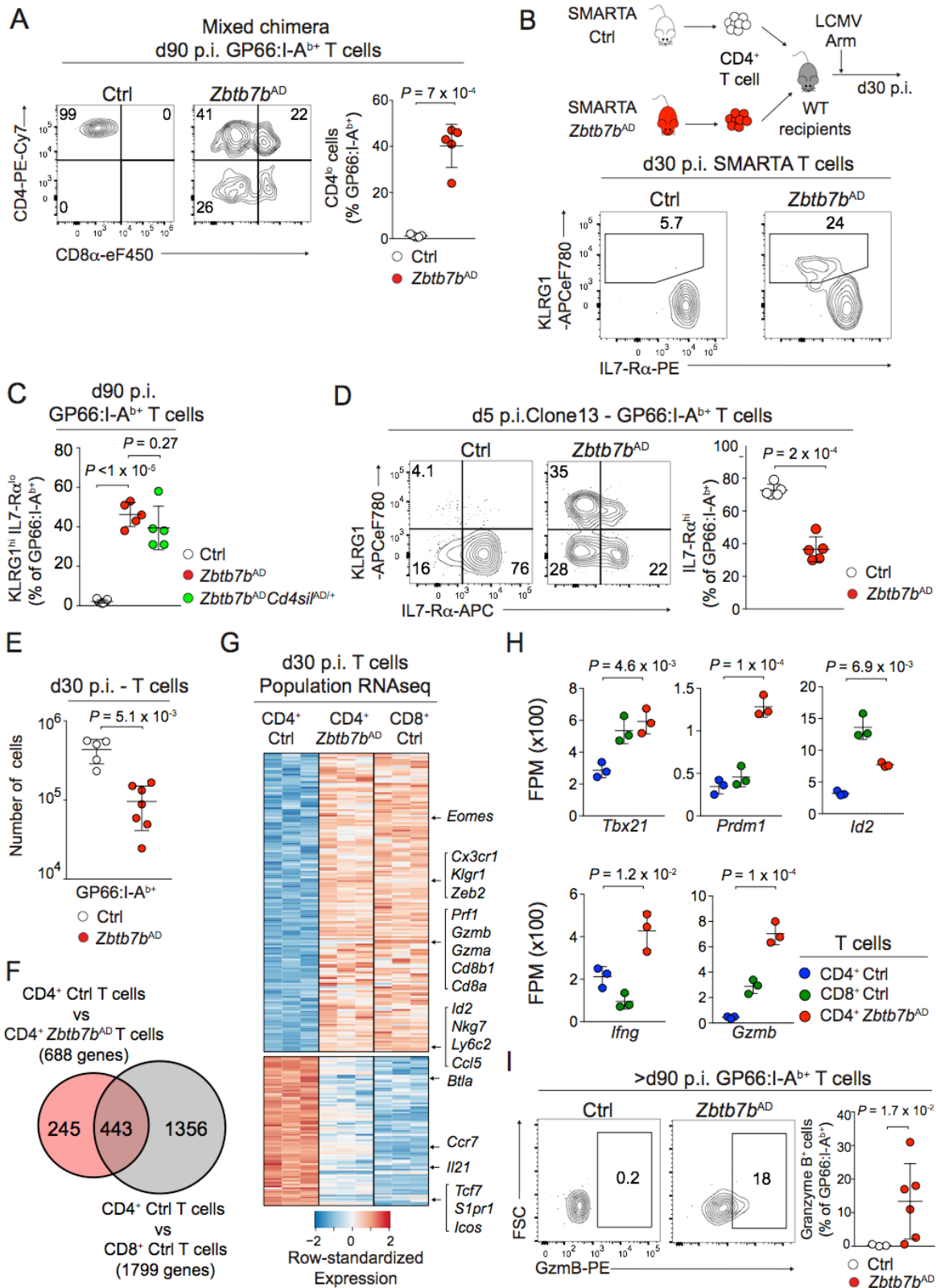


Figure S3: Phenotype and gene expression of Thpok-deficient memory CD4⁺ T cells.

(A) *Zbtb7b*^{AD}:WT or Ctrl:WT mixed bone marrow chimeras were infected with LCMV Arm and analyzed at d90 p.i. Left panels show the expression of CD4 and CD8α on GP66:I-A^{b+} spleen T cells of the indicated genotype. Numbers indicate the percentage of cells in the box. Right panel shows the percentage of CD4^{lo} cells within the GP66:I-A^{b+} T cell population for each genotype (n=5 per group). P-values from unpaired two-sided Welch's *t*-test are indicated.

(B) Naive *Zbtb7b*^{AD} and control SMARTA TCR transgenic cells were adoptively transferred into CD45 congenic WT recipients one day before LCMV Arm infection. Flow cytometry plots show IL-7R α and KLRG1 expression on adoptively transferred cells of the indicated genotype isolated from the spleen at d30 p.i. The percentages of cells in the indicated gates are shown.

(C) Percentage of KLRG1^{hi} IL-7R α ^{lo} cells within the GP66:I-A^{b+} T cell population in *Zbtb7b*^{AD} (n=5), *Zbtb7b*^{AD} *Cd4sil*^{AD} (n=5) and control (n=9) mice analyzed at d90 p.i. with LCMV Arm. P-values from unpaired two-sided Welch's *t*-test are indicated. Data are from one experiment representative of two independent experiments.

(D) *Zbtb7b*^{AD} or control mice were infected with LCMV Arm and re-challenged more than 90 d p.i. with LCMV Clone 13. Left panels show the expression of KLRG1 and IL-7R α on GP66:I-A^{b+} spleen T cells of the indicated genotype. Numbers indicate the percentage of cells in the box. Right panel shows the percentage of IL-7R α ^{hi} cells among GP66:I-A^{b+} T cells of each genotype (n=5 per group). P-value from unpaired two-sided Welch's *t*-test is indicated. Data are representative of two independent experiments.

(E) Numbers of spleen GP66:I-A^{b+} T cells in *Zbtb7b*^{AD} (n=7) and control (n=5) mice 30 d p.i. with LCMV Arm. P-value from unpaired two-sided Welch's *t*-test is indicated. Data are representative of two independent experiments.

(F-H) RNAseq analyses of gene expression at d30 p.i. with LCMV Arm was performed on sorted populations of GP66:I-A^{b+} spleen T cells from *Zbtb7b*^{AD} (CD4 *Zbtb7b*^{AD}) or control (CD4 Ctrl) mice, and of GP33:H-2D^{b+} spleen T cells from WT animals (CD8 Ctrl).

(F) Venn diagrams of the numbers of overlapping genes differentially expressed between CD4 *Zbtb7b*^{AD} vs CD4 Ctrl T cells and CD4 Ctrl vs CD8 Ctrl (adjusted p-value <0.01, log2 fold change > 1).

(G) Genes differentially expressed between CD4 *Zbtb7b*^{AD} and CD4 Ctrl and also differentially expressed between CD4 Ctrl and CD8 Ctrl as shown in (F).

(H) Graphs show RNAseq-determined expression (fragments per million, FPM) of selected genes in all three populations; P-values are from one-way ANOVA. Each symbol represents a biological replicate.

(I) Left panels show Granzyme B expression on GP66:I-A^{b+} spleen T cells of the indicated genotype. Right panel shows the percentage of Granzyme B expressing cells from *Zbtb7b*^{AD} (n=6) and control mice (n=3) more than 90 days p.i. LCMV Arm. P-value is from unpaired two-sided Welch's *t*-test. Data are representative of two independent experiments.

Supplemental Figure 4, Related to Figure 4

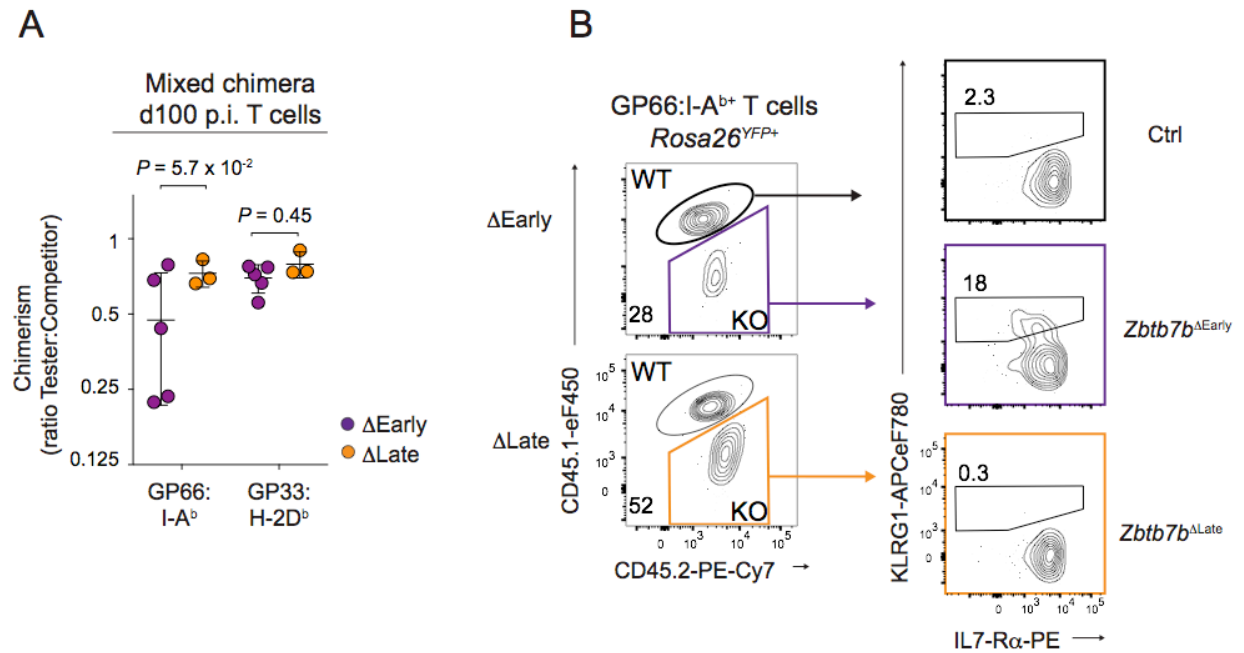


Figure S4: Early requirement for Thpok in the differentiation of memory CD4⁺ T cells.

Mixed bone marrow chimeras (*Cd4-creER^{T2}* *Zbtb7b^{fl/fl}* tester: *Cd4-creER^{T2}* *Zbtb7b^{+/+}* competitor) were infected with LCMV Arm after (Δ Early) or 3 weeks before (Δ Late) tamoxifen treatment.

(A) Graph shows tester-competitor chimerism among spleen GP66:I-A^b *Rosa26^{YFP}* and GP33:H-2D^b T cells at d100 p.i. (Δ Early, n=5; Δ Late, n=3). P-values from two-way ANOVA are indicated.

(B) Right panels show the expression of IL-7R α and KLRG1 on the GP66:I-A^b cell populations defined in the left panels. The percentages of cells in the indicated gates are shown. Data are representative of 2 independent mixed bone marrow experiments and 3 non-chimeric experiments.

Supplemental Figure 5, Related to Figure 5

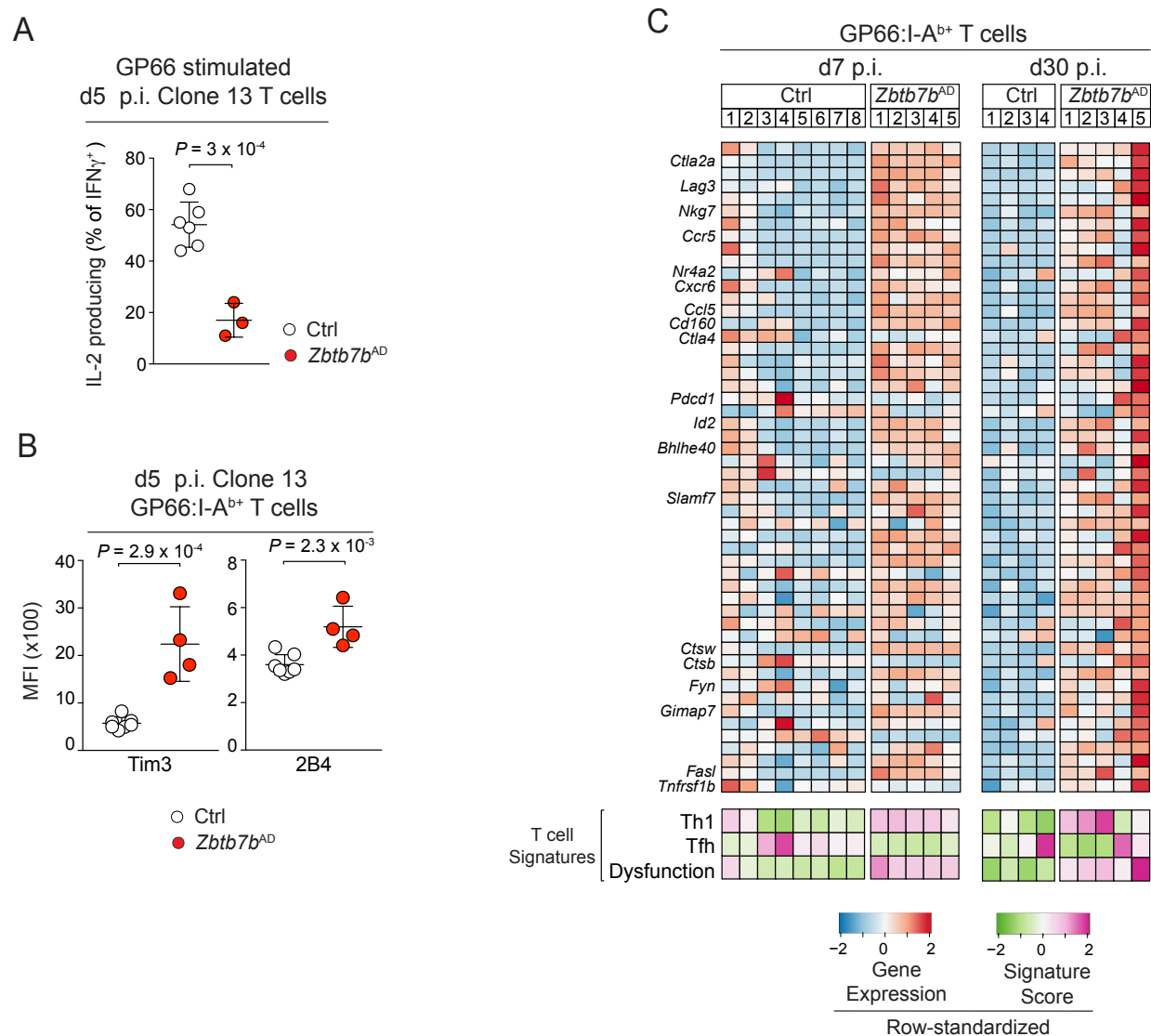


Figure S5: Thpok represses a dysfunctional program in CD4⁺ T cells.

(A, B) *Zbtb7b*^{AD} or control mice were infected with LCMV Arm and re-challenged more than 90 p.i. with LCMV Clone 13. (A) Graph shows the percentage of IL-2 producing among IFN γ -producing *Zbtb7b*^{AD} (n=3) or *Ctrl* (n=6) spleen T cells after GP66 peptide stimulation at d5 post-rechallenge.

(B) Graph shows Tim3 and 2B4 surface expression (mean fluorescence intensities, MFI) on spleen GP66:I-A^b T cells of the indicated genotype 5 days after re-challenge. (A, B) P-values from unpaired two-sided Welch's *t*-test are indicated. Data are representative of two independent experiments.

(C) The "Dysfunction signature" was defined on Cluster d30 *Zbtb7b*^{AD}5 (Fig. 5C) compared to all other d30 clusters. Heatmap shows signature scores and relative expression of dysfunction signature genes across clusters of d7 and d30 cells (as defined respectively in Fig. 2A, B and Fig. 5B, C).

Supplemental Figure 6, Related to Figure 6

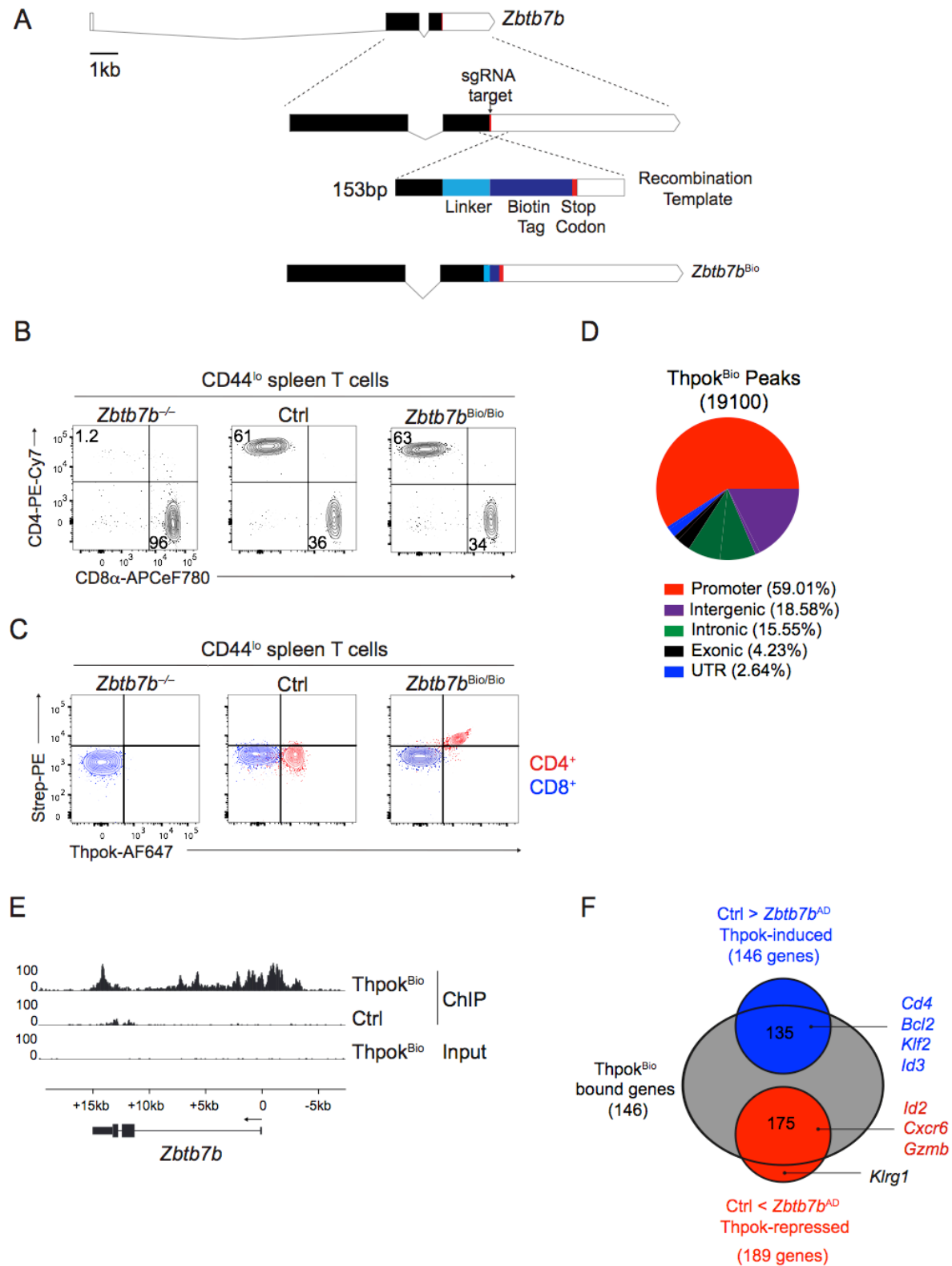


Figure S6: Generation and characterization of *Zbtb7b*^{Bio} mice.

(A) Schematic representation of the generation of the *Zbtb7b*^{Bio} allele. (B, C) *Zbtb7b*^{Bio} animals were crossed with *Rosa26*^{BirA} mice and *Zbtb7b*^{Bio/Bio} *Rosa26*^{BirA} (*Zbtb7b*^{Bio/Bio}), *Zbtb7b*^{-/-} and *Zbtb7b*^{+/-} *Rosa26*^{BirA} (Ctrl) animals were analyzed. (B) Expression of CD4 and CD8α in the spleen T cell population from the indicated animals. The percentages of cells in the indicated gates are

shown. **(C)** Intra-cellular Thpok and biotinylated Thpok were assessed on spleen T cells by anti-Thpok antibody (x axis) and streptavidin (y-axis). Data are representative of two independent experiments. **(D-F)** Thpok DNA binding was assessed by ChIP-seq in activated CD4⁺ T cells from Thpok^{Bio/+} Rosa26^{BirA+} (Thpok^{Bio}) or Rosa26^{BirA+} (Ctrl) animals. **(D)** Pie chart shows the genome-wide distribution of Thpok binding. **(E)** IGV browser tracks show binding by Thpok^{Bio} molecules on the *Zbtb7b* locus (Thpok^{Bio} ChIP), compared to background signals from streptavidin pull-down on Ctrl cells (Ctrl ChIP), and input DNA signal from Thpok^{Bio} cells. **(F)** Gene occupancy by Thpok molecules was compared to Thpok-dependent genes defined on d7 p.i. GP66:I-A^{b+} T cells by scRNAseq.

Supplemental Figure 7, Related to Figure 7

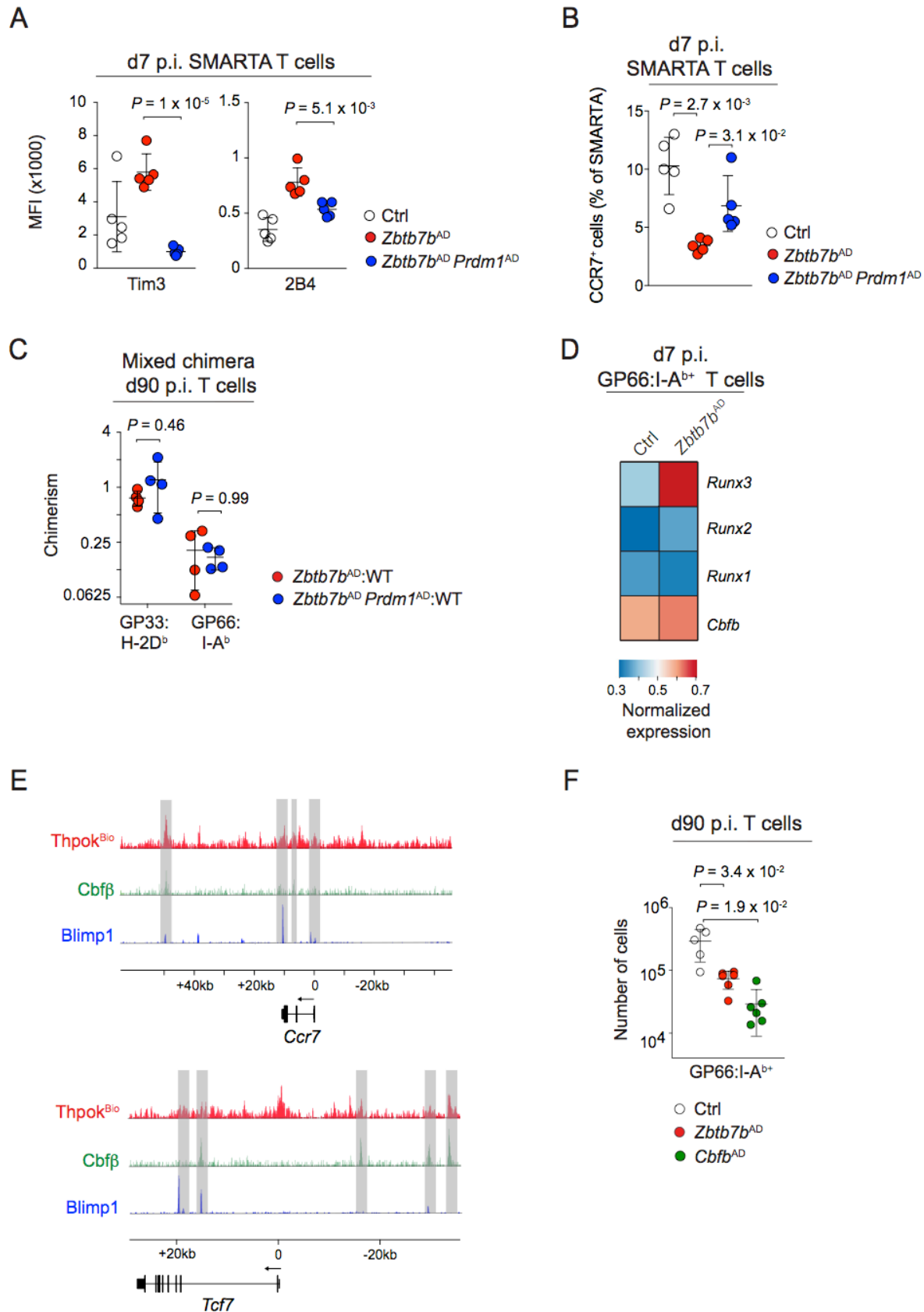


Figure S7: Thpok repression of Blimp1 and Runx3 and memory CD4⁺ T cell differentiation.

(A, B) Naive *Zbtb7b^{AD}*, *Zbtb7b^{AD} Prdm1^{AD}* or control SMARTA cells were adoptively transferred into WT recipients a day before LCMV Arm infection. Graphs show (A) Graphs show Tim3 and 2B4 surface

expression and **(B)** percentage of CCR7 positive cells on the adoptively transferred SMARTA cells of the indicated genotype at d7 p.i. (per group n=5). P-values from unpaired two-sided Welch's *t*-test are indicated. Data are representative of one adoptive transfer experiment and 2 independent experiments on *Zbtb7b*^{AD} or *Zbtb7b*^{AD} *Prdm1*^{AD} and control animals.

(C) *Zbtb7b*^{AD}:WT and *Zbtb7b*^{AD} *Prdm1*^{AD}: WT mixed bone marrow chimeras were analyzed at d90 p.i with LCMV Arm. Graph shows Tester:Competitor ratios among spleen GP33:H-2D^{b+} and GP66:I-A^{b+} T cells in *Zbtb7b*^{AD}:WT (red) and *Zbtb7b*^{AD} *Prdm1*^{AD}:WT (empty circles) chimeras (n=4 per group). Data are representative of one mixed bone marrow experiment and 3 non-chimeric experiments *Zbtb7b*^{AD}, *Zbtb7b*^{AD} *Prdm1*^{AD} and control animals.

(D) Heatmap shows the expression of Runx-related genes on GP66:I-A^{b+} T cells measured by scRNAseq at d7 p.i. with LCMV Arm.

(E) IGV browser tracks show ChIP-seq binding of Thpok, Cbfb and Blimp1 on the *Ccr7* and *Tcf7* loci. Grey shading highlights detected Thpok peaks overlapping with detected CBFβ or Blimp1 peaks

(F) *Zbtb7b*^{AD}, *Cbfb*^{AD} or control mice were infected with LCMV Arm. Graph shows the number of spleen GP66:I-A^{b+} cells in *Zbtb7b*^{AD} (n=6), *Cbfb*^{AD} (n=6) and control (n=5) animals at d90 p.i. P-values from unpaired two-sided Welch's *t*-test are indicated. Data are representative of 2 independent experiments.

Supplemental Table 1, Related to Figures 1, 2 and 5

Cell designation	Capture Session Number	Figure Panel	Timepoint	Cell Type	Genotype	Number of cell capture	Number of cells after filtering	Median Number UMI per cell	Median Number genes per cell
d7 Replicate 1	1	Fig. 1B-D Fig. S1A, D, E	d7 p.i.	CD44 ⁺ T cells	WT	7911	7006	3982	1446
d7 Replicate 2	2	Fig. 1D Fig. S1C-E	d7 p.i.	CD44 ⁺ T cells	WT	2416	2185	5039	1808
d7 GP66:I-A ^{b+} Ctrl	2	Fig. 2A, B Fig. S5C	d7 p.i.	GP66:I-A ^{b+} <i>Rosa26</i> ^{YFP+} T cells	<i>Tfnrsf4-cre</i> ⁺ <i>Zbtb7b</i> ^{+/+}	2459	2154	5277	1758
d7 GP66:I-A ^{b+} <i>Zbtb7b</i> ^{AD}	2	Fig. 2A, B Fig. S5C	d7 p.i.	GP66:I-A ^{b+} <i>Rosa26</i> ^{YFP+} T cells	<i>Tfnrsf4-cre</i> ⁺ <i>Zbtb7b</i> ^{fl/fl}	1414	1330	5889	1984
d30 GP66:I-A ^{b+} Ctrl	2	Fig. 5B, C Fig. S5C	d30 p.i.	GP66:I-A ^{b+} <i>Rosa26</i> ^{YFP+} T cells	<i>Tfnrsf4-cre</i> ⁺ <i>Zbtb7b</i> ^{+/+}	1355	723	2017	832
d30 GP66:I-A ^{b+} <i>Zbtb7b</i> ^{AD}	2	Fig. 5B, C Fig. S5C	d30 p.i.	GP66:I-A ^{b+} <i>Rosa26</i> ^{YFP+} T cells	<i>Tfnrsf4-cre</i> ⁺ <i>Zbtb7b</i> ^{fl/fl}	827	703	2385	1030
d7 GP66:I-A ^{b+} WT	3	Fig. 1E-G Fig S1E-I	d7 p.i.	GP66:I-A ^{b+} T cells	WT	2717	2213	5620	1936

Table S1: scRNAseq captures and metrics

Supplemental Table 2, Related to Figure 1

Signature:	Th1	Th1 (continued)	Tfh	Tcmp	Tmem	Treg	Dysfuntion
	4930453N24Rik	Il18rap	2310001H17Rik	1810058I24Rik	Btg1	Bmyc	2010111I01Rik
	Abrac1	Il2rb	Aldoa	Abli1	Ccnd2	Btg1	Arf6ip1
	Acot7	Itga4	Angptl2	Acot7	Eif4a2	Capg	AW112010
	Actb	Itgb2	Asap1	Acp5	Gm10073	Ccnd2	Bhlhe40
	Actg1	Itgb7	Batf	Actn1	Il7r	Cd2	Ccl3
	Actr3	Klrc1	BC021614	Adgre5	Malat1	Cd74	Ccl4
	Adgre5	Krtcap2	Bcl2a1b	Adk	Rabac1	Ctla4	Ccl5
	Agpat4	Lfng	Borcs8	Ahnak	Rpl12	Foxp3	Ccr5
	Ahnak	Lgals1	Cd160	Anp32a	Rpl13	Gbp7	Cd160
	Anxa1	Lgals3	Cd200	Anxa2	Rpl13-ps3	Gimap7	Cd27
	Anxa2	Lsp1	Cd3g	Arl5c	Rpl18a	Gpx4	Cd3g
	Anxa5	Ly6c2	Cd82	Atp1b3	Rpl36	H2-T22	Cd7
	Anxa6	Mrpl33	Cox14	Atp2b1	Rpl36a	Ifngr1	Cst7
	Ap2s1	Ms4a4b	Ctsb	Bcl2	Rpl37a	Ikzf2	Ctla2a
	Aprt	Ms4a6b	Cxcr5	Bin2	Rpl38	Il7r	Ctla4
	Arhgdib	Mtpn	Ddit4	Ccdc28b	Rpl9	Izumo1r	Ctsb
	Arl6ip5	Myl12a	Dennd2d	Ccr7	Rpl9-ps6	Ltb	Ctsw
	Arpc5	Myl6	Eea1	Cd9	Rplp1	Mbnl1	Cxcr6
	Atp1a1	Myo1f	Fam162a	Cdc25b	Rplp2	Peli1	Cyba
	Atp5c1	Myo1g	Fyn	Cdc42se1	Rps14	Rgs1	Dnaj1
	Atp5h	Ndufb7	Gapdh	Cdkn2d	Rps15	Samsn1	Efh2d
	Atp5l	Nkg7	Gdi2	Crip1	Rps15a	Sdf4	Fas1
	AW112010	Nptn	Gimap5	Crip2	Rps16	Sell	Fyn
	Bhlhe40	Ostf1	Gna13	Dock2	Rps19	Seinc3	Gadd45b
	Calm1	Pglyrp1	Gng2	Emb	Rps21	Shisa5	Gimap7
	Capzb	Plac8	Hif1a	Ezr	Rps23	Tnfrsf18	Gpr65
	Ccl5	Plek	Hmgb1	Fam177a	Rps24	Tnfrsf4	Gzmk
	Ccr2	Podn1	Icos	Fam65b	Rps27		H2-K1
	Cd2	Ppib	Ifi2712a	Glud1	Rps27rt		Hspa1a
	Cd47	Ppp1ca	Isg15	Hint1	Rps28		Hspa8
	Cd48	Ppp3ca	Izumo1r	Id3	Rps29		Hsph1
	Cd52	Prelid1	Limd2	Il6ra	Rps5		Id2
	Cdc42	Psmb3	Lpp	Iqgap1	Rps9		Lag3
	Cdc42ep3	Pycard	Lmp	Itgb1			Lilr4b
	Cfl1	Rap1b	Maf	Itk			Nkg7
	Clic1	Rasgrp2	Marcks1	Jund			Nr4a2
	Cltla	Rbm3	Matk	Klf2			Pdcd1
	Coro1a	Reep5	Mif	Klf3			Plac8
	Cox17	Rnf138	Mmd	Klf6			Psmb10
	Cox5a	Rnf166	Nsg2	mt-Co1			Psmb8
	Cox5b	Rora	Nt5e	Pbxip1			Rgs1
	Crip1	Rpa2	P2rx7	Pdlim1			Rnaset2b
	Crot	Runx3	Pdcd1	Prr13			Samsn1
	Ctla2a	S100a10	Pfkl	Psmad6			Serpina3g
	Ctsc	S100a11	Pfkl	Raf1			Sh2d2a
	Ctsd	S100a13	Pgam1	Rasa3			Shisa5
	Ctsw	S100a4	Pkm	Rasgrp2			Slamf7
	Cxcr6	S100a6	Ppp1r14b	Rora			Sub1
	Cyba	S1pr4	Prkca	Rpa1			Tapbpl
	Cyth4	Sec61b	Ptp4a2	S100a10			Tnfrsf1b
	Dnajc15	Sec61g	Ptpn11	S100a11			Traf4
	Dok2	Selpg	Ptpcap	S100a4			Ubb
	Ech1	Sept11	Pth1	S100a6			
	Emp3	Serpinb6b	Rab37	Sfr1			
	Eno1	Serpinb9	Rgs10	Slamf6			
	Epsti1	Sh3bgrl3	Rnaset2a	Spn			
	Esyt1	Slamf1	Rnaset2b	Srp1			
	Fam107b	Sms	Rpsa	Stk24			
	Gabara12	Spn	Scd2	Stk38			
	Gapdh	St3gal6	Sept7	Tagap			
	Gbp7	Stmn1	Sh2d1a	Tagln2			
	Ggh	Sub1	Smco4	Tcf7			
	Gimap7	Tagln2	Sostdc1	Tcp11l2			
	Glipr2	Taldo1	Tbc1d4	Tspan13			
	Glrx	Tbx21	Tnfrsf8	Tuba1a			
	Gm4950	Tceb2	Tnfrsf8	Vim			
	Gmfg	Thy1	Tox	Vsir			
	Gna15	Tmed2	Tox2	Xm2			
	Gramd3	Tmsb4x	Tpi1				
	Gzmb	Tpm4	Trim8				
	H2afy	Tpst2	Zap70				
	H2afz	Tspo	Zfp361l				
	Hmgb2	Txn1					
	Hsp90b1	Txndc5					
	Id2	Ube2g2					
	Idh3a	Vim					
	Ifngr1	Zyx					
	Il18r1						

Table S2: T cell gene signatures